RVS UNIVERSITY

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Estd. under Andhra Pradesh Private Universities (Establishment and Regulation) Act, 2016

NAAC A+ GRADE ACCREDITED UNIVERSITY

R V S Nagar, Chittoor – 517 127 (AP)

RVSU - B. TECH ACADEMIC REGULATIONS (R26)

COURSE STRUCTURE AND DETAILED SYLLABI

COMPUTER SCIENCE AND BUSINESS SYSTEMS

B. Tech Regular (Full - Time) Four-Year Degree Program

(For the batches admitted from the Academic Year 2026-27)

and

B.Tech. (Lateral Entry Scheme)

(For the batches admitted from the Academic Year 2027-28)

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

SCHOOL OF COMPUTING AND INFORMATION SCIENCES (SCIS)

Website: www.rvsu.org **E-mail**: vicechancellor@rvsu.org

FOREWORD

The Academic Regulations and Syllabi of RVS University, are founded on a commitment to academic excellence, ethical values, innovation and holistic development. Our academic philosophy places the learner at the centre of the educational process, fostering knowledge, professional competence, intellectual curiosity, creativity, character and social responsibility.

The University is firmly committed to Outcome-Based Education (OBE), with clearly articulated learning outcomes and measurable competencies that equip graduates to address the evolving challenges of society, industry and the global knowledge economy.

The curricula integrate disciplinary knowledge with critical thinking, experiential and interdisciplinary learning, research, emerging technologies, entrepreneurship and real-world problem solving. The University seeks to promote academic freedom with accountability, encouraging intellectual inquiry while upholding academic standards and institutional responsibility.

These Academic Regulations establish a transparent and rigorous framework for curriculum design, teaching-learning, assessment, academic progression and governance. The syllabi are conceived as dynamic academic instruments, subject to continuous review and enrichment in response to advances in knowledge, professional requirements, societal needs, national priorities and global developments.

We believe that education must extend beyond employability to develop capable, ethical, innovative and socially responsible professionals and citizens. I am confident that these Academic Regulations and Syllabi will provide a strong foundation for cultivating a culture of quality, academic excellence, innovation, continuous improvement and meaningful contribution to nation-building.

Dr. Ravuri Venkataswamy
Chancellor

VISION, MISSION AND OBJECTIVES

VISION:

To be recognized as world class University that promotes advanced research, contemporary teaching, experiential learning and extension.

MISSION:

- Advance student knowledge through research integrated with education to benefit and address the diversity.
- Undertake challenging problems in multi-disciplinary environments through developing students in domain and soft skills.
- Imbibe a culture with human values and professionalism to develop as creative resources for the nation.
- Establish an ecosystem for faculty and staff for continuous quality improvement.
- Collaborate with industry and premiere educational institutions for quality research and innovation.

OBJECTIVES:

- Establish Centres of excellence in teaching, training and learning.
- Create facilities for quality research which contributes for institutional and national development.
- Developing an ecosystem for innovation and entrepreneurship.
- Collaborate with non-government agencies and industry to undertake extension and social responsibility.

CORE VALUES:

- **Excellence:** Commitment to continuous improvement and innovation in Teaching, Learning, research and extension.
- **Ethics:** Pledge to honesty, integrity, mutual respect, transparency and accountability.
- **Diversity:** Respect to all aspects of diversity.
- **Student First:** Student centric planning and development.
- **Collaboration:** Stakeholder partnership for holistic institutional development and synergistic growth.

RVSU - B. Tech Academic Regulations (R26) (Regular-Fulltime)

(Effective for the students admitted into I year from the Academic Year **2026-27** onwards)

RVS University offers Regular full-time (Four-Year) Bachelor of Technology (B. Tech.) Degree programme under Choice Based Credit System (CBCS) in various branches of Engineering and Technology.

The RVS University shall confer B. Tech Undergraduate degree to candidates who are admitted to the Bachelor of Technology Program and fulfill all the requirements for the award of the degree.

1. **Applicability:**
   All the rules specified herein, approved by the Academic Council, shall be in the force and applicable to the students admitted from the Academic Year 2026-2027 onwards. Any reference to "University" in these rules and regulations stands for RVS University.
2. **Extent:**
   All the rules and regulations, specified hereinafter shall be read as a whole for the purpose of interpretation. As and when a doubt arises, the interpretation of the Vice Chancellor shall be final and ratified by the Academic Council in the forthcoming meeting. As per the requirements of statutory bodies, Vice Chancellor shall be the Chairman, Academic Council.
3. **Award of the Degree**
   a) **Award of the B.Tech. Degree / B.Tech. Degree with a Minor if he/she fulfils the following:**
      i. Pursues a course of study for not less than four academic years and not more than eight academic years. However, for the students availing Gap year facility this period shall be extended by two years at the most and these two years would in addition to the maximum period permitted for graduation (Eight years).
      ii. Registers for 169 and secures all 169 credits

   b) **Award of B.Tech. degree with Honors if he/she fulfils the following:**
      i. Student secures additional 18 credits fulfilling all the requisites of a B.Tech. program i.e., 169 credits.
      ii. Registering for Honors is optional.
      iii. Honors is to be completed simultaneously with B.Tech. programme.

   c) **Award of B.Tech. degree with Minors**
      i. Student secures additional 18 credits fulfilling all the requisites of a B.Tech. program i.e., 169 credits.
      ii. Registering for Minors is optional.
      iii. Minors is to be completed simultaneously with B.Tech. programme
4. Students, who fail to fulfil all the academic requirements for the award of the degree within eight academic years from the year of their admission, shall forfeit their seat in B.Tech. course and their admission stands cancelled. This clause shall be read along with clause 3 a) i).
5. **Admissions**
   Admission to the B. Tech Program shall be made subject to the eligibility, qualifications and specialization prescribed by the A.P. State Government / RVS University from time to time. Admissions shall be made either based on the merit rank obtained by the student in the common entrance examination conducted by the A.P. Government and / or RVS University or any other order of merit approved by the A.P. Government, subject to reservations as prescribed by the Government / RVS University from time to time.
6. **Program related terms**
   Credit: A unit by which the course work is measured. It determines the number of hours of instruction required per week. One credit is equivalent to one hour of teaching (Lecture / Tutorial) or two hours of practical work / field work per week.

Credit Definition:

1Hr. Lecture(L) per week	1 Credit
1Hr.Tutorial (T) per week	1 Credit
1Hr.Practical (P) per week	0.5 Credit
2Hrs.Practical (Lab) per week	1 Credit

a) ***Academic Year:*** Two consecutive (one odd + one even) semesters constitute one academic year.

b) ***Choice Based Credit System (CBCS):*** The CBCS provides a choice for students to select from the prescribed courses.

7. Semester/Credits:

i) A semester comprises of 15 weeks and an academic year is divided into two semesters.

ii) The summer term is for eight weeks during summer vacation. Internship / apprenticeship / work-based vocational education and training can be carried out during the summer term, especially by students who wish to exit after two semesters or four semesters of study.

iii) Regular courses may also be completed well in advance through MOOCs satisfying prerequisites.

8. Structure of the Undergraduate Programme:

All courses offered for the undergraduate program (B.Tech.) are broadly classified as follows:

S. No.	Category	Break-up of Credits (Total 169)	Percentage of total Credits	AICTE Recommendation (%)
1.	Humanities and Social Science including Management (HS)	15	9%	8 – 9%
2.	Basic Sciences (BS)	22	13 %	12 -16%
3.	Engineering Sciences (ES)	24	15%	10– 18%
4.	Professional Core (PC)	57	33 %	30– 36%
5.	Electives – Professional (PE) & Open (OE); Domain Specific Skill Enhancement Courses (SEC)	35	20 %	19 -23%
6.	Internships & Project work (PR)	16	10 %	8 –11%
7.	Mandatory Courses (MC)	Non-credit	Non-credit	-

The total degree credits become 169, including 2 credits for online Examination. The syllabus for this online examination shall comprise core courses studied by the student up to IV B. Tech, I Semester and this examination will be conducted along with IV B. Tech, I semester Regular Examinations.

9. Course Classification:

All subjects / courses offered for the undergraduate programme in Engineering & Technology (B.Tech. degree programmes) are broadly classified as follows:

S. No.	Broad Course Classification	Course Category	Description
1.	Foundation Courses	Foundation courses	Includes Mathematics, Physics and Chemistry; fundamental engineering courses; humanities, social sciences and management courses
2.	Core Courses	Professional Core Courses (PC)	Includes subjects related to the parent discipline / department / branch of

			Engineering
3.	Elective Courses	Professional Elective Courses (PE)	Includes elective subjects related to the parent discipline / department / branch of Engineering
		Open Elective Courses (OE)	Elective subjects which include interdisciplinary subjects or subjects in an area outside the parent discipline / department / branch of Engineering
		Domain specific skill enhancement courses (SEC)	Interdisciplinary / job-oriented / domain courses which are relevant to the industry
4.	Project & Internships	Project	B.Tech. Project or Major Project
		Internships	Summer Internships–Community based and Industry Internships; Industry oriented Full Semester Internship
5.	Mandatory Courses	Mandatory non-credit courses	Covering subjects of developing desired attitude among the learners
6.	Audit Courses	Non-Mandatory non-credit courses	Covering subjects which develop additional skills and values among the learners

10. Programme Pattern

i. Total duration of B. Tech (Regular) Programme is four academic years.
ii. Each academic year of study is divided into two semesters.
iii. Minimum number of instructional weeks in each semester is 15 weeks.
iv. There shall be mandatory student induction program for fresher's, with a three-week duration before the commencement of first semester. Physical activity, Creative Arts, Universal Human Values, Literary, Proficiency Modules, Lectures by Eminent People, Visits to local Areas, Familiarization to Dept./Branch & Innovations etc., are included as per the guidelines issued by AICTE.
v. Health / wellness / yoga / sports / NSS / NCC / Scouts & Guides / Community service activities are made mandatory as credit courses for all the undergraduate students.
vi. Courses like Environmental Sciences, Indian Knowledge System, Cybersecurity Awareness for Engineers, Gender Sensitization, Technical Paper Writing & IPR are offered as non-credit mandatory courses for all the undergraduate students. Audit Courses are also introduced as non-Mandatory non-Credit courses.
vii. Design Thinking & Tinkering Labs are made mandatory as credit courses for all the undergraduate students.
viii. A comprehensive online examination will be conducted in IV Year – I Semester for 2 credits
ix. Increased flexibility for students through an increase in the elective component of the curriculum, with 05 Professional Elective courses and 04 Open Elective courses.
x. Professional Elective Courses, include the elective courses relevant to the chosen specialization / branch. Proper choice of professional elective courses can lead to students specializing in emerging areas within the chosen field of study.
xi. A total of 04 Open Electives is offered in the curriculum.
xii. While choosing the electives, students shall ensure that they do not opt for the courses with syllabus contents similar to courses already pursued.
xiii. A pool of interdisciplinary / job-oriented / domain skill courses which are relevant to the industry are integrated into the curriculum of all disciplines. There shall be 05 skill-oriented courses offered during II B. Tech I Semester to IV B. Tech I Semester. Among the five skill courses, four courses shall focus on the basic and advanced skills related to the domain/interdisciplinary courses and the other shall be a soft skills course.
xiv. Students shall undergo mandatory summer internships, for a minimum of six-eight weeks duration at the end of second and third year of the programme. The internship at the end of second year shall be community oriented and industry internship at the end of third year.
xv. There shall also be mandatory full internship in the final semester of the

programme along with the Major project work.

xvi. Undergraduate degree with Honors and Minors is introduced by the University for the students having good academic record.

xvii. The Dean shall take measures to implement Virtual Labs (https://www.vlab.co.in) which provide remote access to labs in various disciplines of Engineering and will help student in learning basic and advanced concepts through remote experimentation. Student shall be made to work on virtual lab experiments during the regular labs.

xviii. The Dean shall assign a faculty advisor / mentor after admission to a group of students from same department to provide guidance in courses registration / career growth / placements / opportunities for higher studies / GATE / other competitive exams.

xix. Preferably 25% of course work for the theory courses in every semester shall be conducted in the blended mode of learning.

11. Evaluation Process

The performance of a student in each semester shall be evaluated subject wise with a maximum of 100 marks for theory and 100 marks for practical subject. Summer Internships shall be evaluated for 50 marks, Full Internship in final semester shall be evaluated for 50 marks & Project work in final semester shall be evaluated for 200 marks, mandatory courses with no credits shall be evaluated for 40 marks through mid-semester examinations.

A student has to secure not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the mid semester and end examination marks taken together for the theory, practical, design, drawing subject or project etc. In case of a mandatory course, he / she should secure 40% of the total marks.

Theory Courses

Assessment Method	Marks
Continuous Internal Assessment	40
Semester End Examination	60
Total	**100**

i) For theory subject, the distribution shall be 40 marks for Internal Evaluation and 60 marks for the End-Examination.

ii) For practical subject, the distribution shall be 40 marks for Internal Evaluation and 60 marks for the End-Examination.

iii) If any course contains two different branch subjects, the syllabus shall be written in two parts with 3 units each (Part-A and Part-B) and external examination question paper shall be set with two parts each for 30 marks.

iv) If any subject is having both theory and practical components, they will be evaluated separately as theory subject and practical subject. However, the marks scored in both theory and practical components shall be added and the same may be condensed to 100 marks for calculation of grade point.

a) Continuous Internal Evaluation

i) For theory subjects, during the semester, there shall be two mid-term examinations. Each mid-term examination shall be evaluated for 40 marks of which 10 marks for objective paper (20 minutes duration), 25 marks for subjective paper (90 minutes duration) and 5 marks for assignment.

ii) Objective paper shall contain for 05 short answer questions with 1 mark each or maximum of 10 multiple choice questions for 05 marks. Subjective paper shall contain 3 either or type questions (totally six questions from 1 to 6) of which student has to answer one from each either-or type of questions. Each question carries 10 marks. The total marks for the subjective paper will be 30 marks.

Note:

- The objective paper shall be prepared in line with the quality of competitive examinations questions.
- The subjective paper shall contain 3 either or type questions of equal

weightage of 10 marks. Any fraction shall be rounded off to the next higher mark.

- The objective paper examination shall be conducted along with the subjective paper examination.
- Assignments shall be in the form of problems, mini projects, design problems, slip tests, case studies, quizzes or any other relevant and suitable activity prescribed by the department., depending on the course content. The first assignment for the first two units and the second assignment for the next three units and each assignment shall be evaluated for 5 marks. The average of two assignments shall be considered for assignment marks.

iii) If the student is absent for the mid semester examination, no re-exam shall be conducted and mid semester marks for that examination shall be considered as zero.

iv) First midterm examination shall be conducted for I, II units of syllabus with one either or type question from each unit and third either or type question from both the units. The second midterm examination shall be conducted for III, IV and V units with one either or type question from each unit.

v) Final mid semester marks shall be arrived at by considering the marks secured by the student in both the mid examinations with 80% weightage given to the better mid exam and 20% to the other.

For Example:

Marks obtained in first mid: 25 Marks
obtained in second mid: 20
Final mid semester Marks: (25x0.8) + (20x0.2) =24

If the student is absent for any one midterm examination, the final mid semester marks shall be arrived at by considering 80% weight age to the marks secured by the student in the appeared examination and zero to the other. For Example:

Marks obtained in first mid: Absent Marks
obtained in second mid:25
Final mid semester Marks: (25x0.8) +(0x0.2) =20

b) End Examination Evaluation:

End examination of theory subjects shall have the following pattern:

i) There shall be 6 questions and all questions are compulsory.

ii) Question I shall contain 5 compulsory short answer questions for a total of 10 marks such that each question carries 2 marks. There shall be 1 short answer question from each unit.

iii) The questions from 2 to 6 shall be set by covering one unit of the syllabus for each question. In each of the questions from 2 to 6, there shall be either / or type questions of 10 marks each. Student shall answer any one from each either-or type questions.

End examination of theory subjects consisting of two parts of different subjects, for Example: Basic Electrical & Electronics Engineering shall have the following pattern:

i) Question paper shall be in two parts viz., Part-A and Part-B with equal weightage of 30 marks each.

ii) In each part, question 1 shall contain 6 compulsory short answer questions for a total of 6 marks such that each question carries 1mark.

iii) In each part, questions from 2 to 4, there shall be either / or type questions of 8 marks each. Student shall answer any one from each either-or type questions.

iv) The questions from 2 to 4 shall be set by covering one unit of the syllabus for each question.

c) For practical courses, there shall be a continuous evaluation during the semester for 40 sessional marks and end examination shall be for 60 marks.

i) Day-to-day work in the laboratory shall be evaluated for 20 marks by the concerned laboratory teacher based on the record / viva and 20 marks for the internal test.

ii) The end examination shall be evaluated for 60 marks, conducted by the concerned laboratory teacher and a senior faculty member in the subject from the same department.

a. Procedure: 20marks
b. Experimental work & Results: 30 marks
c. Viva-voce: 10marks.

In a practical subject consisting of two parts (Eg: Basic Electrical & Electronics Engineering Lab), the end examination shall be conducted for 60 marks as a single laboratory in 3 hours. Mid semester examination shall be evaluated as above for 40 marks in each part and final mid semester marks shall be arrived by considering the average of marks obtained in two parts.

iii) For the subject having design and/or drawing, such as Engineering Drawing, the distribution of marks shall be 40 for mid semester evaluation and 60 for end examination.

Assessment Method	Marks
Continuous Internal Assessment	40
Semester End Examination	60
Total	**100**

Day-to-day work shall be evaluated for 10 marks by the concerned subject teacher based on the reports/submissions prepared in the class. And there shall be two mid-term examinations in a semester for duration of 2 hours each for 30 marks with weightage of 80% to better mid marks and 20% for the other. The subjective paper shall contain 3 either or type questions of equal weightage of 10 marks. There shall be no objective paper in mid semester examination. The sum of day-to-day evaluation and the mid semester marks will be the final sessional marks for the subject.

The end examination pattern for Engineering Graphics, shall consists of 5 questions, either/or type, of 12 marks each. There shall be no objective type questions in the end examination. However, the end examination pattern for other subjects related to design/drawing, multiple branches, etc is mentioned along with the syllabus.

iv) There shall be no external examination for mandatory courses with zero credits. However, attendance shall be considered while calculating aggregate attendance and student shall be declared to have passed the mandatory course only when he/she secures 40% or more in the internal examinations. In case, the student fails, a re-examination shall be conducted for failed candidates for 40 marks satisfying the conditions mentioned in item 1 & 2 of the regulations.

v) The laboratory records and mid semester test papers shall be preserved for a minimum of 3 years in the concerned Department as per the University norms and shall be produced to the Committees of the University as and when the same are asked for.

Note:

- The end-semester examinations shall be conducted based on the theoretical components of the prescribed syllabus.
- Assignments shall be designed to incorporate AI-integrated elements of the syllabus to promote applied and experiential learning.
- The mid-semester examinations may include a combination of both theoretical content and AI-integrated components

d) For theory and practical combined courses, the theory and practical components shall be evaluated for 100 marks each. The evaluation for theory component shall be conducted in line with theory courses. The evaluation for practical component shall be conducted in line with practical courses. However, to declare pass in the course, the student has to satisfy minimum academic requirements (pass marks) in both theory and laboratory components as stipulated for theory and laboratory courses. The marks scored in theory component and practical component shall be interpolated to 50 marks each separately and these interpolated marks in theory component (50 marks) and practical component (50 marks) shall be combined for awarding of grade in the concerned course.

12. Skill oriented Courses

i) There shall be five skill- oriented courses offered during II B. Tech I semester to IV B. Tech I semester.

ii) Out of the five skill courses two shall be skill-oriented courses from the same domain. Of the remaining three skill courses, one shall be a soft skill course and the remaining two shall be skill-advanced courses from the same domain / Interdisciplinary / Job oriented.

iii) The course shall carry100 marks and shall be evaluated through continuous assessments during the semester for 40 sessional marks and end examination shall be for 60 marks. Day-to-day work in the class / laboratory shall be evaluated for 40 marks by the concerned teacher based on the regularity / assignments / viva/ mid semester test. The end examination similar to practical examination pattern shall be conducted by the concerned teacher and an expert in the subject nominated by the Dean.

iv) The Head of the Department shall identify a faculty member as coordinator for the course. A committee consisting of the Head of the Department, coordinator and a senior Faculty member nominated by the Head of the Department shall monitor the evaluation process. The marks/grades shall be assigned to the students by the above committee based on their performance.

v) The student shall be given an option to choose either the skill courses being offered by the School or to choose a certificate course being offered by industries/Professional bodies or any other accredited bodies. If a student chooses to take a Certificate Course offered by external agencies, the credits shall be awarded to the student upon producing the Course Completion Certificate from the agency. A committee shall be formed at the level of the School to evaluate the grades/marks given for a course by external agencies and convert to the equivalent marks/grades.

vi) The recommended courses offered by external agencies, conversions and appropriate grades/marks are to be approved by the University at the beginning of the semester. The Dean of the School shall forward such proposals to the University for approval.

vii) If a student prefers to take a certificate course offered by external agency, the department shall mark attendance of the student for the remaining courses in that semester excluding the skill course in all the calculations of mandatory attendance requirements upon producing a valid certificate as approved by the University.

13. Massive Open Online Courses (MOOCs):

To promote self-learning and lifelong learning, students are encouraged to enroll in Massive Open Online Courses (MOOCs) offered through **SWAYAM, SWAYAM Plus, and NPTEL** or other University-approved platforms.

A student may earn up to 40% of the total credits prescribed in a semester through MOOCs, subject to the University's guidelines and equivalence requirements.

Further, every student shall mandatorily complete atleast one MOOC course before the commencement of the IV Year II Semester. In addition, the Fifth Professional Elective (PE-V) and the Fourth Open Elective (OE-IV) shall be mandatorily completed through the SWAYAM/SWAYAM Plus platforms. This provision is intended to facilitate students in undertaking the Long-Term Internship during the final semester. Accordingly, every student shall mandatorily complete a total of three MOOC courses for the award of the B.Tech. degree. The following are guidelines for MOOC courses.

i. Students may earn a maximum of 40% of the credits in a semester through MOOCs for PE, OE, and SEC. Further, students of III and IV year are also permitted to pursue Core Courses (CC) through MOOCs.
ii. All the courses for Honors and Minors may be studied through MOOCs platform.
iii. Students shall register for MOOCs courses of a minimum duration of either 8 weeks or 12 weeks with the prior approval of the Head of the Department. Credits shall be awarded as follows:
   a) 2 credits for an 8-week course
   b) 3 credits for a 12-week course
iv. The NCrF rating for the Course is 4.5 for UG program
v. The Head of the Department shall nominate a faculty mentor to monitor the academic progress of students enrolled in MOOCs. It is also responsibility of the

mentor to see that the registered subjects are in the approved list provided by the university.

vi. Students shall have the option to choose either the regular offline (classroom) mode or the SWAYAM mode for a particular subject. However, a student shall not be permitted to opt for both modes simultaneously for the same subject. Students opting for SWAYAM shall be exempted from attending regular classroom sessions for that subject, and attendance shall not be counted. Further, the Departments should not insist the student to opt SWAYAM course compulsorily.

vii. All students in a class are not required to opt for the same SWAYAM course. Departments shall arrange regular classes for students opting for offline mode, while the same faculty member may act as a mentor for students enrolled in the SWAYAM course. If all students opt for a SWAYAM course, the subject need not be included in the timetable; however, a mentor shall be assigned to address student queries as per mutual convenience.

viii. Students who completed the prescribed assignments successfully and secured at least 40% of the internal marks shall be permitted to appear for the end-semester examination conducted either by the National Testing Agency (NTA)/NPTEL or by the University. University conducted examinations shall be held along with regular end-semester examinations, with a weightage of 60% for the end-term examination and 40% for assignments and quizzes evaluated by the SWAYAM Course Coordinator.

ix. Credits shall be awarded only upon submission of a successful course completion certificate. Internal marks obtained through SWAYAM shall be extrapolated to 40 and external marks interpolated to 60, and shall exactly correspond to the marks secured by the student in the SWAYAM course.

x. In case where University prescribed passing marks are higher than SWAYAM's minimum requirements, the University norms shall prevail. In such cases, even if a student submits a SWAYAM certificate but fails to secure the University's minimum passing marks, the student shall be declared unsuccessful and must appear for the supplementary examination.

xi. In cases where a student obtains less than the prescribed minimum internal marks (16 out of 40) in a SWAYAM course, and the said course is not offered again in the subsequent semester, the University may, through a special notification, conduct a full course supplementary examination for 100 marks during the final semester of the programme. The examination shall be conducted in accordance with the University question paper pattern for **60 marks**, strictly following the University prescribed syllabus and it will be extrapolated to 100.

xii. For SWAYAM courses where end-term examinations are conducted by the University, students shall not be issued a SWAYAM certificate.

xiii. If a SWAYAM course carries more credits than the corresponding RVSU course, credit transfer shall be restricted to the number of credits prescribed by RVSU to maintain uniformity in total credit requirements. Credit transfer shall not be permitted if the SWAYAM course carries fewer credits than the RVSU course.

Students who have qualified in the proctored examinations conducted through MOOCs platform can apply for credit transfer as specified and are exempted from appearing internal as well as external examination (for the specified equivalent credit course only) conducted by the University.

Necessary amendments in rules and regulations regarding adoption of MOOC courses would be proposed from time to time.

14. **Credit Transfer Policy**

Adoption of MOOCs is mandatory, to enable Blended model of teaching-learning as also envisaged in the NEP2020. As per University Grants Commission (Credit Frame work for Online Learning Courses through SWAYAM) Regulation, 2016, the University shall allow up to a maximum of 40% of the credits in a semester being offered in a particular programme i.e., via Swayam/Swayam Plus courses through MOOCs platform.

i) The University shall offer credit mobility for MOOCs and give the equivalent credit weightage to the students for the credits earned through online learning courses.
ii) Student registration for the MOOCs shall be only through the respective department, it is mandatory for the student to share necessary information with the department.
iii) Credit transfer policy will be applicable to the Professional & Open Elective courses only.
iv) The concerned department shall identify the courses permitted forced it transfer.
v) The University/Departments shall notify at the beginning of semester the list of the online learning courses eligible for credit transfer.
vi) The Departments shall designate a faculty member as a Mentor for each course to guide the students from registration till completion of the credit course.
vii) The University shall ensure no overlap of MOOC exams with that of the University examination schedule. In case of delay in results, the University will re-issue the marks sheet for such students.
viii) Student pursuing courses under MOOCs shall acquire the required credits only after successful completion of the course and submitting a certificate issued by the competent authority along with the percentage of marks and grades.
ix) The Departments shall submit the following to the examination section of the university:
    a) List of students who have passed MOOC courses in the current semester along with the certificate of completion.
    b) Undertaking form filled by the students for credit transfer.
x) The University shall resolve any issues that may arise in the implementation of this policy from time to time and shall review its credit transfer policy in the light of periodic changes brought by UGC, SWAYAM, NPTEL and State Government.

15. Academic Bank of Credits (ABC)

The University is implementing Academic Bank of Credits (ABC) to promote flexibility in curriculum as per NEP 2020 to

i. Provide option of mobility for learners across the universities of their choice.
ii. Provide option to gain the credits through MOOCs from approved digital platforms.
iii. Facilitate award of certificate / diploma / degree in line with the accumulated credits in ABC.
iv. Execute Multiple Entry and Exit system with credit count, credit transfer and credit acceptance from students' account.

16. Mandatory Internships

Summer Internships: Two summer internships with a minimum of 06-08 weeks duration (2 credits each), done at the end of second and third years, respectively are mandatory. It shall be completed in collaboration with local industries, Govt. Organizations, construction agencies, Power projects, software MNCs or any industries in the areas of concerned specialization of the Undergraduate (B. Tech) program. One of the two summer internships at the end of second year (Community Service Project) shall be society oriented and shall be completed in collaboration with government organizations / NGOs & others. The other internship at the end of third year is Industry Internship and shall be completed in collaboration with Industries. The student shall register for the internship as per course structure after commencement of academic year. The guidelines issued by the APSCHE / University shall be followed for carrying out and evaluation of Community Service Project and Industry Internship.

Evaluation of the summer internships shall be through the departmental committee. A student will be required to submit a summer internship report to the concerned department and appear for an oral presentation before the departmental committee comprising of Head of the Department, Mentor of the internship and a senior faculty member of the department. A certificate of successful completion from industry shall be included in the report. The report and the oral presentation shall carry 50% weightage each. It shall be evaluated for 50 external marks. There shall be no internal marks for Summer Internship. A student shall secure minimum 40% of marks for successful completion. In case, if a student fails, he/she shall reappear as and when semester supplementary examinations are conducted by the University.

Full Semester Internship and Project work:
In the final semester, the student should mandatorily register and undergo internship and in parallel he / she should work on a project with well-defined objectives. At the end of the semester the candidate shall submit an internship completion certificate and a project report. A student shall also be permitted to submit project report on the work carried out during the internship.

The Long-Term Industry Internship undertaken in the final semester (IV Year – II Semester) shall be evaluated for 50 marks by a committee consisting of the Head of the Department along with two senior faculty members of the department. Further, every student shall undertake at least one Industry Visit between the III B. Tech I Semester and IV B. Tech I Semester, as part of the mandatory internships and practical training requirements.

The project report shall be evaluated with an external examiner. The total marks for project work 200 marks (8 credits) and distribution shall be 80 marks for internal and 120 marks for external evaluation. The supervisor assesses the student for 40 marks (Report: 20 marks, Seminar: 20 marks). At the end of the semester, all projects shall be showcased at the department for the benefit of all students and staff and the same is to be evaluated by the departmental Project Review Committee consisting of supervisor, a senior faculty and HOD for 40 marks. The external evaluation of Project Work is a Viva-Voce Examination conducted in the presence of internal examiner (Supervisor), Head of the department and external examiner appointed by the University and is evaluated for 120 marks.

The Departments shall facilitate and monitor the student internship programs. Completion of internships is mandatory, if any student fails to complete internship, he/she will not be eligible for the award of degree. In such cases, the student shall repeat and complete the internship.

17. Student Development Program (SDP) – Department-Wise Workshop (Mandatory Non-Credit Course)

Each Department shall design and conduct a **Student Development Programme (SDP) / Department-Wise Workshop** relevant to its discipline for all B. Tech. students. The programme shall carry **Zero Credits** and shall be conducted **at the end of the III B. Tech. I Semester**.

The following guidelines shall govern the conduct and evaluation of the programme:

i. **Programme Planning:** The Head of the Department (HoD) shall prepare and notify the workshop schedule, syllabus/content, learning outcomes, and assessment plan at least **two weeks prior to the commencement** of the programme.
ii. **Attendance:** Attendance in all workshop sessions shall be **mandatory**. Students shall attend the programme as per the notified schedule.
iii. **Evaluation:** The programme shall be evaluated for a total of **50 marks**, comprising:
    - **25 marks** for continuous assessment based on participation, assignments, practical activities, presentations, or other evaluation components prescribed by the Department; and
    - **25 marks** for a final examination to be conducted on the **last day of the workshop**.
iv. **Pass Requirements:** A student shall be declared to have **successfully completed** the programme only if he/she secures 40% marks in both continuous assessment and final examination taken together:
v. **Submission of Marks:** The HoD shall submit the finalized marks list to the **Controller of Examinations, RVSU,** in the prescribed format within the timeline notified by the University.
vi. **Certification:** A **Workshop Completion Certificate** shall be issued only to those students who have successfully fulfilled the prescribed attendance and pass requirements.
vii. **Supplementary Examination:** A student who fails to secure the minimum prescribed marks shall be required to appear for the **supplementary examination** conducted by the University in the subsequent semester, as per the University rules

and schedule.

Successful completion of the **Student Development Programme (SDP)/Department-Wise Workshop** is **mandatory for the award of the B.Tech. degree**, and its completion status shall be reflected in the student's academic records.

18. Guidelines for offering a Minor Degree

To promote interdisciplinary knowledge among the students, the students admitted into B. Tech. in a major stream/branch are eligible to obtain degree in Minor in another stream.

i) Minor Degree is introduced in the curriculum of all B. Tech. programs offering a major degree and is applicable to all B. Tech (Regular and Lateral Entry) students admitted in Engineering & Technology.

ii) A student shall earn additional 18 credits (comprising 5 theory courses and 2 laboratory / project courses, offered in tracks) for award of B. Tech. (Minor degree). This is in addition to the credits essential for obtaining the Undergraduate (B. Tech) degree in Major Discipline (i.e., 169 credits).

iii) A student is permitted to register for Minor degree in II B. Tech II semester after the results of II B. Tech I Semester are declared and students may be allowed to take maximum two subjects per semester pertaining to the Minor Degree from II B. Tech II Semester onwards.

iv) The Dean of the School shall arrange separate class work and timetable of the courses offered under Minor Degree program.

v) Courses that are used to fulfil the student's primary major (Open Elective) may not be double counted towards the Minor Degree. Courses with content substantially equivalent to courses in the student's primary Major Degree may not be counted towards the Minor Degree.

vi) Students can complete the courses offered under Minor Degree either in the University or in online platforms like SWAYAM with a minimum duration of 12 weeks for a 3-credit course and 8 weeks duration for a 2-credit course satisfying the criteria for credit mobility. If the courses under Minor Degree are offered in conventional mode, then the teaching and evaluation procedure shall be similar to regular B. Tech courses.

vii) The attendance for the registered courses under Minor Degree and regular courses offered for Major degree in a semester are to be considered separately.

viii) A student shall maintain an attendance of 75% in all registered courses under Minor Degree to be eligible for attending semester end examinations.

ix) A student registered for Minor Degree shall pass in all subjects that constitute the requirement for the Minor degree program. No class/division (i.e., second class, first class and distinction, etc.) shall be awarded for Minor degree programme.

x) If a student drops or is terminated from the Minor degree program, the additional credits so far earned cannot be converted into open or core electives; they will remain extra. However, such students will receive a separate grade sheet mentioning the additional courses completed by them.

xi) The Minor Degree will be mentioned in the degree certificate. Every Department in the School shall offer Minor Degree program and A Minor Degree course on Quantum Computing shall be offered for all branches, with the curriculum framed by the CSE Department.

xii) A student who fails in any course of the Minor Degree Programme shall be permitted to appear for the corresponding supplementary examination(s) as per the University rules. However, the student shall successfully complete all the prescribed courses of Minor Degree program before Completion of Major Degree Programme. Failure to do so shall result in the cancellation of admission to the Minor Degree Programme, and no Minor credential shall be awarded.

xiii) All the courses offered under Minor Degree may also be pursued through MOOCs. The Bloom's Taxonomy level for Minor courses shall be 3 and below.

19. Guidelines for offering Honors Degree

The objective of introducing B. Tech. (Honors) is to facilitate the students to choose additionally the specialized courses of their choice and build their competence in a specialized area in the UG level. The programme is a best choice for academically excellent students having good academic record and interest towards higher studies and research.

i. Honors Degree is introduced in the curriculum of all B. Tech. programs offering a major degree and is applicable to all B. Tech (Regular and Lateral Entry) students admitted in Engineering & Technology.

ii. Students with CGPA of 7 and no back logs upto II B. Tech I semester are eligible to register for Honors Degree programs.

iii. A student shall earn additional 18 credits (comprising 5 theory courses and 2 laboratory / project courses, offered in tracks) for award of B. Tech. (Honors) degree from same branch / department / discipline registered for major degree. This is in addition to the credits essential for obtaining the Undergraduate degree in Major Discipline (i.e., 169 credits).

iv. A student is permitted to register for Honors Degree in II B. Tech II Semester after the results of II B. Tech I Semester are declared and students may be allowed to take maximum two subjects per semester pertaining to the Honors Degree from II B. Tech II Semester onwards.

v. The concerned Head of the Departments shall arrange separate class work and time table of the courses offered under Honors Degree program.

vi. Courses that are used to fulfil the student's primary major may not be double counted towards the Honors Degree Courses with content substantially equivalent to courses in the student's primary Major may not be counted towards the Honors Degree.

vii. Students can complete the courses offered under Honors Degree either in the University or in online platforms like SWAYAM with a minimum duration of 12 weeks for a 3-credit course and 8 weeks duration for a 2-credit course satisfying the criteria for credit mobility. If the courses under Honors Degree are offered in conventional mode, then the teaching and evaluation procedure shall be similar to regular B. Tech courses.

viii. The attendance for the registered courses under Honors Degree and regular courses offered for Major degree in a semester are to be considered separately.

ix. A student shall maintain an attendance of 75% in all registered courses under Honors Degree to be eligible for attending semester end examinations.

x. A student registered for Honors Degree shall pass in all subjects that constitute the requirement for the Honors degree program. No class/division (i.e., second class, first class and distinction, etc.) shall be awarded for Honors degree programme.

xi. Registration for Honors Degree program continue even if the student fails in any of the regular course after his registration to Honors Degree Course.

xii. A student who fails in any course of the Honor Degree Programme shall not be permitted to write the supplementary examination and the registration for the Honors Degree Programme will be cancelled.

xiii. If a student drops or is terminated from the Honors Degree program, the additional credits so far earned cannot be converted into open or core electives; they will remain extra. However, such students will receive a separate grade sheet mentioning the additional courses completed by them.

xiv. The Honors Degree will be mentioned in the degree certificate as Bachelor of Technology (Honors) in XYZ. For example, B.Tech. (Honors) in Mechanical Engineering.

xv. All the courses offered under Honors Degree may also be pursued through MOOCs. The Bloom's Taxonomy level for Honors Degree courses shall be 3 and above

Enrolment into Honors Degree:

i) Students of a Department/Discipline are eligible to opt for Honors Degree program offered by the same Department/Discipline

ii) The enrolment of student into Honors Degree is based on the CGPA obtained in the major degree program. CGPA shall be taken up to II B. Tech I semester in case of regular entry students and only II B. Tech I semester in case of lateral entry students. Students having 7 CGPA without any backlog subjects will be permitted to register for Honors Degree.

iii) If a student is detained due to lack of attendance either in Major Degree or in Honors Degree, registration for Honors Degree shall be cancelled.

iv) Transfer of credits from Honors Degree to regular B. Tech degree and vice-versa shall not be permitted.

v) Honors Degree is to be completed simultaneously with a Major degree program.

Registration for Honors / Minors:

i) The eligible and interested students shall apply through the HOD of his/her parent department. The whole process should be completed within one week before the start of every semester. Selected students shall be permitted to register the courses under Honors/Minors Degree.

ii) The students enrolled in the Honors/Minors Degree courses will be monitored continuously. An advisor/mentor from parent department shall be assigned to a group of students to monitor the progress.

iii) There shall be prescribed fee for registration of subjects for Honors/Minors Degree program offered in offline at the University.

Failure & Supplementary Examination Procedure for Honors/Minor Degree Courses:

i) Time Limit for Completion:
The student must successfully complete all Honors/ Minor Degree courses within the regular period of study applicable to the B. Tech. programme.

ii) Cancellation of Honors/ Minor Degree Registration:
If the student fails to complete all Honors / Minor Degree requirements within the stipulated regular period of study, the Honors/ Minor Degree registration shall stand cancelled, and the additional credits earned shall remain extra. These credits should not be counted for the award of the major Degree.

20. Attendance Requirements:

i) A student shall be eligible to appear for the University external examinations if he/she acquires a minimum of 40% attendance in each subject and 75% of attendance in aggregate of all the subjects. b) Condonation of shortage of attendance in aggregate up to 10% (65% and above and below 75%) in each semester may be granted by the Academic Council of the University.

ii) Shortage of Attendance below 65% in aggregate shall in NO CASE be condoned.

iii) A stipulated fee shall be payable towards condonation of shortage of attendance to the University.

iv) Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examination of that class and their registration shall stand cancelled.

v) A student will not be promoted to the next semester unless he satisfies the attendance requirements of the present semester. They may seek readmission for that semester from the date of commencement of class work.

vi) If any candidate fulfils the attendance requirement in the present semester, he shall not be eligible for readmission into the same class.

vii) If the learning is carried out in blended mode (both offline & online), then the total attendance of the student shall be calculated considering the offline and online attendance of the student.

viii) For induction programme attendance shall be maintained as per AICTE norms.

ix) Attendance in SWAYAM courses shall be excluded from the calculation of the student's overall attendance requirement for the semester.

21. Conduct of Semester End Examination and Evaluation:

i) Semester end examination shall be conducted by the Controller of Examination (COE) by inviting Question Papers from the External / Internal Subject Experts. Director of Evaluation will decide the External / Internal subject experts.

ii) The answer papers of semester end examination should be evaluated externally / internally.

iii) The marks for the internal evaluation components will be added to the external evaluation marks secured in the Semester – End examinations, to arrive at total marks for any subject in that semester.

iv) Performance in all the subjects is tabulated program-wise and will be scrutinized by the office of the Controller of Examinations. Total marks obtained in each subject are converted into letter grades. Finally subject-wise marks and grades details,

subject-wise and branch-wise pass percentages are calculated.

22. Results Committee

i) Results Committee comprising of Vice Chancellor/nominee of Vice Chancellor, Director of Evaluation, Controller of Examinations, Dean of the School and One Senior Professor nominated by the Vice Chancellor on rotation basis will oversee the details of marks, grades and pass percentages of all the subjects and branch-wise pass percentages.

ii) Office of the Controller of Examinations will generate student-wise result sheets and the same will be published through University website.

iii) Student-wise Grade Sheets are generated and issued to the students.

23. Promotion Rules:

The following academic requirements must be satisfied in addition to the attendance requirements mentioned in section 20.

i) A student shall be promoted from first year to second year if he/she fulfils the minimum attendance requirement as per university norms.

ii) A student will be promoted from II B. Tech to III B. Tech if he/she fulfils the academic requirement of securing 40% of the credits (any ***decimal*** fraction should be ***rounded off*** to ***lower*** digit) up to in the subjects that have been studied up to II B. Tech I Semester.

iii) A student shall be promoted from III B. Tech to IV B. Tech if he/she fulfils the academic requirements of securing 40% of the credits (any ***decimal*** fraction should be ***rounded off*** to ***lower*** digit) in the subjects that have been studied up to III B. Tech I Semester.

And in case a student is detained for want of credits for a particular academic year by ii) & iii) above, the student may make up the credits through supplementary examinations and only after securing the required credits he/she shall be permitted to join in the III B. Tech I Semester or IV B. Tech I Semester respectively as the case may be.

iv) When a student is detained due to lack of credits/shortage of attendance he/she may be re-admitted when the semester is offered after fulfilment of academic regulations. In such case, he/she shall be in the Academic Regulations into which he/she is readmitted.

24. Grading:

As a measure of the student's performance, a 10-point Absolute Grading System using the following Letter Grades and corresponding percentage of marks shall be followed:
After each course is evaluated for 100 marks, the marks obtained in each course will be converted to a corresponding letter grade as given below, depending on the range in which the marks obtained by the student fall.

Structure of Grading of Academic Performance

Range in which the marks in the subject fall	Grade	Grade points Assigned
90 &above	S (Superior)	10
80-89	A (Excellent)	9
70-79	B (Very Good)	8
60-69	C (Good)	7
50-59	D (Average)	6
40-49	E (Below Average)	5
< 40	F (Fail)	0
Absent	Ab (Absent)	0

i) A student obtaining Grade 'F' or Grade 'Ab' in a subject shall be considered failed and will be required to reappear for that subject when it is offered the next supplementary examination.

ii) For non-credit audit courses, "Satisfactory" or "Unsatisfactory" shall be indicated instead of the letter grade and this will not be counted for the computation of

SGPA/CGPA/Percentage.

Computation of Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA):

The Semester Grade Point Average (SGPA) is the ratio of sum of the product of the number of credits with the grade point scored by a student in all the courses taken by a student and the sum of the number of credits of all the courses undergone by a student, i.e.,

$$SGPA=\Sigma(C_i \times G_i)/\Sigma C_i$$

Where, "C_i" is the number of credits of the i^{th} subject and "G_i" is the grade point scored by the student in the i^{th} course.

The Cumulative Grade Point Average (CGPA) will be computed in the same manner considering all the courses undergone by a student overall the semesters of a program, i.e.,

$$CGPA=\Sigma(C_i \times S_i)/\Sigma C_i$$

Where "S_i" is the SGPA of the i^{th} semester and "C_i" is the total number of credits up to that semester.

Both SGPA and CGPA shall be rounded off to 2 decimal points and reported in the Grade Sheets / transcripts.

While computing the SGPA the subjects in which the student is awarded Zero grade points will also be included.

Grade Point: It is a numerical weightage allotted to each letter grade on a 10-point scale.

Letter Grade: It is an index of the performance of students in a said course. Grades are denoted by the letters S, A, B, C, D, E and F.

25. Grade sheet:

A grade sheet (Marks Memorandum) will be issued to each student Indicating his performance in all subjects registered in that semester indicating the SGPA and CGPA. SGPA and CGPA will be rounded off to the second place of decimal.

26. Consolidated Grade Sheet:

After successful completion of the entire Program of study, a Consolidated Grade Sheet containing performance of all semesters will be issued as a final record. Duplicate Consolidated Grade Sheet will also be issued, if required, after payment of requisite fee.

27. Award of Class: After a student has satisfied the requirements prescribed for the completion of the program and is eligible for the award of B. Tech. Degree, he/she shall be placed in one of the following four classes:

Class Awarded	CGPA Secured
First Class with Distinction	≥7.5
First Class	≥6.5<7.5
Second Class	≥5.5 <6.5
Pass Class	≥4.5 <5.5

Both SGPA, CGPA and Marks shall be given in the Grade Sheets.

28. Personal Verification /Recounting / Revaluation / Final Valuation

28.1 Personal Verification of Answer Scripts:

Candidates appear in a particular semester end-examinations may appeal for verification of their answer script(s) for arithmetic correction in totaling of marks and any omission / deletion in evaluation within 7 days from the date of declaration of results at the office of the Controller of Examinations on the prescribed proforma and by paying the prescribed fee per answer script.

It is clarified that personal verification of answer script shall not tantamount to revaluation of answer script. This is only a process of reverification by the candidate. Any mistake / deficiency with regard to arithmetic correction in totaling of marks and any omission / deletion in evaluation if found, the institution will correct the same.

28.2 Recounting / Revaluation:

Students shall be permitted for request for recounting/revaluation of the Semester-End examination answer scripts within a stipulated period after payment of prescribed fee. After recounting or revaluation, records are updated with changes if any and the student will be issued a revised grade sheet. If there are no changes, the same will be intimated to the students.

28.3 Final Valuation:

Students shall be permitted for request for final valuation of the Semester–End Examination answer scripts within a stipulated period after the publication of the revaluation results by paying the necessary fee. The final valuation shall be carried out by an expert not less than Associate Professor as per the scheme of valuation supplied by the examination branch in the presence of the student, Controller of Examinations and Director of Evaluation. However, students are not permitted to discuss / argue with the examiner. If the increase in marks after final valuation is equal to or more than 15% of the previous valuation marks, the marks obtained after final valuation shall be treated as final. If the variation of marks after final valuation is less than 15% of the previous valuation marks, then the earlier valuation marks shall be treated as the final marks.

29. With–holding of Results:

If the candidate has any case of indiscipline pending against him, the result of the candidate shall be withheld, and he will not be allowed/promoted into the next higher semester. The issue of degree is liable to be withheld in such cases.

30. Multiple Entry / Exit Option

(a) Exit Policy:

The students can choose to exit the four-year programme at the end of first / second / third year.

i) **UG Certificate in (Field of study/discipline)** - Programme duration: First year (first two semesters) of the undergraduate (B. Tech) programme, followed by an additional exit 10-credit bridge course(s) lasting two months, including at least 6-credit job-specific internship / apprenticeship that would help the candidates acquire job-ready competencies required to enter the work force.

ii) **UG Diploma (in Field of study/discipline)** - Programme duration: First two years (first four semesters) of the undergraduate (B. Tech) programme, followed by an additional exit 10-credit bridge course(s) lasting two months, including at least 6-creditjob-specific internship / apprenticeship that would help the candidates acquire job-ready competencies required to enter the workforce.

iii) **Bachelor of Science (in Field of study / discipline)** i.e., **B. Sc. Engineering in (Field of study / discipline)**- Programme duration: First three years (first six semesters) of the undergraduate (B. Tech) programme.

(b) Entry Policy:

Modalities on multiple entry by the student into the B. Tech. programme will be provided in due course of time.

Note:

The University shall resolve any issues that may arise in the implementation of Multiple Entry and Exit policies from time to time and shall review the policies in the light of periodic changes brought by UGC, AICTE and State Government.

31. Gap Year Concept:

Gap year concept for Student Entrepreneur in Residence is introduced and outstanding students who wish to pursue entrepreneurship/ become entrepreneur are allowed to take a break of one year at any time after II B. Tech to pursue full-time entrepreneurship programme / to establish start-ups. This period may be extended to two years at the most and these two years would not be counted for the time for the maximum time for graduation. The Head of the respective Department shall forward such proposals submitted by the students to the University. An evaluation committee constituted by the University shall evaluate the proposal submitted by the student and the committee shall decide whether to permit the student(s) to avail the Gap Year or not.

32. Transitory Regulations

Discontinued, detained, or failed candidates are eligible for readmission as and when the semester is offered after fulfilment of Academic Regulations. Candidates who have been detained for want of attendance or not fulfilled academic requirements or who have failed after having undergone the course in earlier regulations or have discontinued and wish to continue the course are eligible for admission into the unfinished semester from the date of commencement of classwork with the same or equivalent subjects as and when subjects are offered, and they will follow the academic regulations into which they are readmitted.

Candidates who are permitted to avail Gap Year shall be eligible for rejoining into the succeeding year of their B. Tech from the date of commencement of class work, and they will follow the academic regulations into which they are readmitted

33. Minimum Instructional Days for a Semester:

The minimum instructional weeks excluding exams for each semester shall be 15 weeks.

34. Medium of Instruction

The medium of instruction of the entire B. Tech undergraduate programme in Engineering & Technology (including examinations and project reports) will be in English only.

35. Student Transfers:

Student transfers shall be as per the guidelines issued by the Government of Andhra Pradesh and the University from time to time.

36. Re-registration for Improvement of Internal Evaluation Marks:

A candidate shall be given one chance to re-register for each subject provided the internal marks secured by a candidate are less than 50% and has failed in the end examination.

36.1 The candidate should have completed their four years of course work and obtained examinations results for all the four years and are permitted to re-register as and when the respective subjects are offered.

36.2 Out of the subjects the candidate has failed in the examination due to Internal Evaluation marks secured being less than 50%, the candidate shall be given one chance for each Theory subject and for a maximum of ten Theory subjects for Improvement of Internal evaluation marks.

36.3 The candidate has to re-register for the chosen subjects and fulfill the academic requirements.

36.4 For re-registration the candidates have to submit the applications to the Dean of the Concerned School through the Head of the Department by paying the requisite fees (For each subject, the candidate has to pay a requisite fee and the amount is to be remitted in the form of D.D./ Challan in favour of the Registrar, RVS University) and get approval from the Dean of the Concerned School before the start of the semester in which re-registration is required.

36.5 In the event of availing the Improvement of Internal evaluation marks, the internal evaluation marks as well as the End Examinations marks secured in the previous attempt(s) for the re-registered subjects stand cancelled. The internal marks obtained by the students in the re-registration shall be treated as final.

37. General Instructions:

37.1 The academic regulations should be read as a whole for purpose of any interpretation.

37.2 Malpractices rules nature and punishments are appended

37.3 Where the words "he", "him", "his", occur in the regulations, they also include "she", "her", "hers", respectively.

37.4 The University may change or amend the academic regulations or syllabi at any time and the changes or amendments shall be made applicable to all the students on rolls with effect from the dates notified by the University.

37.5 In the case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Vice-Chancellor is final.

FAILURE TO READ AND UNDERSTAND THE REGULATIONS IS NOT AN EXCUSE

RVSU Academic Regulations-R26 for B. Tech (Lateral Entry Scheme)

(Effective for the students admitted into II year B. Tech through Lateral Entry Scheme from the Academic Year **2027-28** onwards)

1. Award of the Degree

(a) Award of the B. Tech. Degree / B. Tech. Degree with a Minor if he/she fulfils the following:

i. Pursues a course of study for not less than three academic years and not more than six academic years. However, for the students availing Gap year facility this period shall be extended by two years at the most and these two years would in addition to the maximum period permitted for graduation (Six years).

ii) Registers for 127 credits and secures all 127 credits

(b) Award of B. Tech. degree with Honors if he / she fulfils the following:

i) Student secures additional 18 credits fulfilling all the requisites of a B. Tech. program i.e., 127 credits.

ii) Registering for Honors is optional.

iii) Honors is to be completed simultaneously with B.Tech. programme.

(c) Award of B.Tech. degree with Minor

i) Student secures additional 18 credits fulfilling all the requisites of a B.Tech. program i.e., 127 credits.

ii) Registering for Minor Degree is optional.

iii) Minor Degree is to be completed simultaneously with B. Tech. programme.

2. Students, who fail to fulfil the requirement for the award of the degree within six consecutive academic years from the year of admission, shall forfeit their seat.

3. Course Pattern

i) The entire course of study is three academic years on semester pattern.

ii) A student eligible to appear for the end examination in a subject but absent at it or has failed in the end examination may appear for that subject at the next supplementary examination offered.

iii) When a student is detained due to lack of credits / shortage of attendance the student may be re-admitted when the semester is offered after fulfilment of academic regulations, the student shall be in the academic regulations into which he/she is readmitted.

4. All other academic regulations, including those relating to attendance, examinations, promotion, evaluation, and the award of the degree, shall be the same as those prescribed for regular B. Tech. students, unless otherwise specified

Identification of Courses

B. Tech

Each course shall be uniquely identified by an alphanumeric code of width 7 characters as given below.

No. of Digits	Description
First two digits	Year of regulations Ex:26
Next one letter	**Type of program:** A: B. Tech B: M. Tech C: M.B.A D: M.C.A E: BBA F: BCA G: B. Sc (Hon) H: B. Sc (AHS)
Next two letters	**Code of program**: CE: Civil Engineering EE: Electrical and Electronics engineering ME: Mechanical Engineering EC: Electronics and Communications Engineering CS: Computer Science and Engineering CD: Computer Science and Engineering (Data Science) CM: Computer Science and Engineering (Artificial Intelligence and Machine Learning) IT: Information Technology CA: Computer Science and Engineering (Artificial Intelligence) CC: Computer Science and Engineering (Cyber Security) CO: Computer Science and Engineering (IOT) CB: Computer Science and Business Systems AI: Artificial Intelligence AD: Artificial Intelligence and Data Science CQ: Computer Science and Engineering (Quantum Computing) DE: Computer Science and Engineering (Artificial Intelligence and Data Engineering) AR: Computer Science and Engineering (Artificial Intelligence and Robotics) AC: Computer Science and Engineering (Cloud Computing and Automation) DT: Electronics and Communication Engineering with Specialization in Drone Technology
Last two digits	Indicate serial numbers: ≥ 01

Ex:

26ACE01; 26AEE01; 26AME01; 26AEC01; 26ACS01; 26ACD01; 26ACM01; 26AIT01; 26ACA01; 26ACC01; 26ACO01; 26ACB01; 26AAI01; 26AAD01; 26ACQ01; 26ADE01; 26AAR01; 26AAC01; 26ADT01; 26AHS01; 26AMB01

RULES FOR DISCIPLINARY ACTION FOR MALPRACTICES / IMPROPER CONDUCT IN EXAMINATIONS

	Nature of Malpractices/Improper conduct	Punishment
	If the candidate:	
1.(a)	Possesses or keeps accessible in examination hall, any paper, notebook, programmable calculators, Cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which he is appearing but has not made use of (material shall include any marks on the body of the candidate which can be used as an aid in the subject of the examination)	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other candidate orally or by any other body language methods or communicates through cell phones with any candidate or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the candidates involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the candidate is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The Hall Ticket of the candidate is to be cancelled and sent to the University.
3.	Impersonates any other candidate in connection with the examination.	The candidate who has impersonated shall be expelled from examination hall. The candidate is also debarred for four consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. The performance of the original candidate who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The candidate is also debarred for four consecutive semesters from class work and all University examinations, if his involvement is established. Otherwise, the candidate is debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.

4.	Smuggles in the Answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
5.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject only.
6.	Refuses to obey the orders of the Chief Superintendent /Assistant - Superintendent /any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer-in charge or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the College campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the candidate(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. If the candidate physically assaults the invigilator/ officer-in-charge of the Examinations, then the candidate is also debarred and forfeits his/her seat. In case of outsiders, they will be handed over to the police and a police case is registered against them.
7.	Leaves the exam hall taking away answer script or intentionally tears of the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
8.	Possess any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat.

9.	If student of the college, who is not a candidate for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Student of the colleges expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat. Person (s) who do not belong to the College will be handed over to police and, a police case will be registered against them.
10.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject only or in that subject and all other subjects the candidate has appeared including practical examinations and project work of that semester / year examinations, depending on the recommendation of the committee.
12	Tampering of the NPTEL / SWAYAM / SWAYAM certificate	Cancellation of student performance in the entire semester
13.	If any malpractice is detected which is not covered in the above clauses 1 to 12 shall be reported to the University for further action to award suitable punishment.	-

Note:-

Whenever the performance of a student is cancelled in any subject/subjects due to Malpractice, he has to register for End Examinations in that subject/subjects consequently and has to fulfil all the norms required for the award of Degree.

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

VISION AND MISSION OF THE DEPARTMENT

VISION:

To emerge as a distinctive centre integrating computing and business intelligence, developing technology-enabled leaders who transform organizations through innovation, analytics and digital solutions.

MISSION:

M1: Integrating student learning with computing, business processes, analytics and digital technologies for technology-enabled management.

M2: Fostering entrepreneurial thinking and innovation in digital business models, intelligent enterprises and technology-led transformation.

M3: Developing workplace readiness through case-based learning, consulting projects, internships and sustained industry engagement.

M4: Enabling responsible technology adoption that enhances organizational performance, sustainability and societal value.

M5: Strengthening faculty expertise at the computing-business interface and providing enterprise software, analytics platforms and collaborative learning environments.

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1 Graduates will pursue advanced education in computing, business analytics, management, information systems and interdisciplinary technological domains.

PEO2 Graduates will excel in professional careers by integrating computing knowledge with business processes, analytics, management practices and digital transformation initiatives.

PEO3 Graduates will identify business opportunities enabled by technology and develop innovative digital products, services and technology-driven enterprises.

PEO4 Graduates will continuously develop technological, managerial and business competencies to respond effectively to changing industry and market requirements.

PROGRAM SPECIFIC OUTCOMES (PSOs)

On successful completion of the Program, the graduates of B. Tech. Computer Science and Business Systems will be able to:

PSO1: Apply principles of computer science, business analytics, management and systems analysis to analyse and solve technology-enabled organizational and business problems.

PSO2: Design, develop, integrate and evaluate software applications, enterprise information systems and digital solutions aligned with business processes, customer requirements and organizational objectives using appropriate computing and software engineering tools.

PSO3: Apply data analytics, business intelligence, cloud computing, enterprise technologies and digital technologies to support data-driven decision-making, process optimization, innovation and digital transformation in business organizations.

PROGRAM OUTCOMES (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

RVS UNIVERSITY

FIND YOUR PLACE | FOLLOW YOUR PASSION

Estd. under Andhra Pradesh Private Universities (Establishment and Regulation) Act, 2016

NAAC A+ GRADE ACCREDITED UNIVERSITY

B.TECH. - COURSE STRUCTURE – R26
(Applicable from the academic year 2026-27 onwards)

INDUCTION PROGRAMME
(0 Semester – 3 weeks duration)

Sl. No.	Activities
1.	Physical Activities - Sports, Yoga and Meditation, Plantation
2.	Career Counselling
3.	Orientation to all branches - Career Options, Tools,etc.
4.	Orientation on admitted branch - Corresponding Labs, Tools and Platforms
5.	Proficiency Modules & Productivity Tools
6.	Assessment on Basic Aptitude and Mathematical Skills
7.	Remedial Training in Foundation Courses
8.	Human Values & Professional Ethics
9.	Communication Skills - focus on Listening, Speaking, Reading, Writing skills
10.	Concepts of Programming
11.	Mentor-Mentee Groups Meet
12.	Visits in Local Area
13.	Lectures & Workshops by Eminent People
14.	Feedback and Report on the Program

Course Structure & Scheme of Examination

I B. Tech I Semester- CSBS **Academic Regulations: R26**

S. No	Category	Course Code	Course Title	Hours / Week			Credits	Scheme of Examination (Maximum Marks)		
				L	T	P		CIA	SEE	Total
1	HS	26AHS01	Communicative English	2	0	2	3	80	120	200
2	BS	26AHS05	Applied Chemistry	3	0	2	4	80	120	200
3	BS	26AHS03	Linear Algebra & Calculus	3	0	0	3	40	60	100
4	PC	26ACS01	Computational Thinking and Problem Solving with Python	3	0	2	4	80	120	200
5	ES	26AAI01	AI Tools and Applications	1	0	4	3	40	60	100
6	ES	26AIT01	IT Workshop	0	0	3	1.5	40	60	100
7	HS	26AHS08	Health and Wellness, Yoga and Sports	0	0	1	0.5	100	-	100
TOTAL				**12**	**0**	**14**	**19**	**460**	**540**	**1000**

I B. Tech II Semester-CSBS **Academic Regulations: R26**

S. No	Category	Course Code	Course Title	Hours / Week			Credits	Scheme of Examination (Maximum Marks)		
				L	T	P		CIA	SEE	Total
1	BS	26AHS06	Applied Physics	3	0	2	4	80	120	200
2	PC	26ACS02	Data Structures Using Python	3	0	2	4	80	120	200
3	BS	26AHS07	Differential Equation and Vector calculus	3	0	0	3	40	60	100
4	PC	26ACA01	Artificial Intelligence	3	0	2	4	80	120	200
5	PC	26ACS04	Computer Organization & Architecture	3	0	0	3	40	60	100
6	PC	26AEC03	Digital Logic Design	3	0	0	3	40	60	100
7	PC	26AAI02	Prompt Engineering	0	0	3	1.5	40	60	100
8	HS	26AHS09	NSS/NCC/ Scouts and Guides	0	0	1	0.5	100	-	100
TOTAL				**18**	**0**	**10**	**23**	**500**	**600**	**1100**

I B. Tech I Semester - Common to CIVIL, EEE, ME, CSE, CSD, CSM, CSE (AI&DE), CSBS and CSE (AI & R)
I B. Tech II Semester - Common to ECE, IT, CAI, CSC, CSO, AI, AI & DS, ECE (DT), CSE (CCA) and CSE (QC)

L	T	P	C
2	0	2	3

26AHS01: COMMUNICATIVE ENGLISH

Course Outcomes (COs):

After successful completion of the course, students will be able to:

CO1 : Identify the context, main idea and specific information from social, academic and transactional dialogues and short audio texts.

CO2 : Apply appropriate grammatical structures, vocabulary related to Engineering and contextual language conventions to construct and edit grammatically correct sentences and comprehend academic and technical texts.

CO3 : Participate effectively in pair and group discussions by organizing ideas, using appropriate discourse markers, and communicating clearly in formal and informal academic contexts.

CO4 : Write clear, grammatically correct and contextually appropriate academic and professional documents, including official letters, technical reports and resumes.

CO5 : Demonstrate fluency, clarity and confidence in academic and professional oral communication through structured presentations, extempore speaking and conversational activities.

UNIT- I: HUMAN VALUES: GIFT OF MAGI (SHORT STORY) 6 Hours

Listening : Identifying the topic, the context and specific pieces of information by listening to short audio texts and answering a series of questions.

Speaking : Asking and answering general questions on familiar topics such as home, family, work, studies and interests and introducing oneself and others.

Reading : Skimming to get the main idea of a text; scanning to look for specific piece of information.

Writing : Mechanics of Writing-Capitalization, Spellings, Punctuation-Parts of Sentences.

Grammar : Parts of Speech, Basic Sentence Structures-forming questions

Vocabulary	:	Synonyms, Antonyms, Affixes (Prefixes/Suffixes), Root words.
AI Intervention	:	Grammarly, Merriam-Webster Vocabulary Building, Gamma.app

UNIT – II: NATURE: THE BROOK BY ALFRED TENNYSON (POEM)

5 Hours

Listening	:	Answering a series of questions about main ideas and supporting ideas after listening to audio text.
Speaking	:	Discussion in pairs / small groups on specific topics followed by short structural talks.
Reading	:	Identifying sequence of ideas; recognizing verbal techniques that help to link the ideas in a paragraph together
Writing	:	Structure of a paragraph - Paragraph writing (specific topics)
Grammar	:	Cohesive devices -linkers, use of articles and zero article; prepositions
Vocabulary	:	Homonyms, Homophones, Homographs.
AI Intervention	:	ELSA Speak, ChatGPT / Google Gemini, Merriam-Webster Vocabulary Building

UNIT – III: BIOGRAPHY: ELON MUSK 7 Hours

Listening	:	Listening for global comprehension and summarizing what is listened to.
Speaking	:	Discussing specific topics in pairs or small groups and reporting what are discussed.
Reading	:	Reading a text in detail by making basic inferences -recognizing and interpretingspecific context clues; strategies to use text clues for comprehension.
Writing	:	Summarizing, Note-making, paraphrasing
Grammar	:	Verbs - tenses; subject-verb agreement; Compound words, Collocations
Vocabulary	:	Compound words, Collocations
AI Intervention	:	Grammarly, ELSA Speak, Speech-to-Text, SmallTalk2Me

UNIT- IV: INSPIRATION: THE TOYS OF PEACE BY SAKI 7 Hours

Listening	:	Making predictions while listening to conversations/ transactional dialogues without video; listening with video
Speaking	:	Role plays for practice of conversational English in academic contexts (formal and informal) - asking for and giving information / directions
Reading	:	Studying the use of graphic elements in texts to convey

		information, reveal trends / patterns / relationships, communicate processes or display complicated data.
Writing	:	Letter Writing: Official Letters, Technical Report, Resumes
Grammar	:	Reporting verbs, Direct & Indirect speech, Active & Passive Voice
Vocabulary	:	Words often confused, Jargons
AI Intervention	:	Real-time grammar, ELSA Speak, Speech-to-Text

UNIT- V: MOTIVATION: THE POWER OF INTRAPERSONAL COMMUNICATION (AN ESSAY) 5 Hours

Listening	:	Identifying key terms, understanding the concepts and answering a series of relevant questions that test comprehension
Speaking	:	Formal oral presentations on topics from academic contexts
Reading	:	Reading comprehension
Writing	:	Critical Writing - Writing structured essays on specific topics.
Grammar	:	Editing short texts –identifying and correcting common errors in grammar and usage (articles, prepositions, tenses, subject verb agreement)
Vocabulary	:	Technical Jargons
AI Intervention	:	Notion AI, ELSA Speak, Speech-to-Text, Gamma.app

LIST OF EXERCISES:

Exercise - 1: Vowels & Consonants
Exercise - 2: Neutralization/Accent Rules
Exercise - 3: Communication Skills & JAM
Exercise - 4: Role Play or Conversational Practice
Exercise - 5: E-mail Writing & Technical Report Writing
Exercise - 6: Cover Letter, Resume Writing & SOP
Exercise - 7: Group Discussions-methods & practice
Exercise - 8: Debates - Methods & Practice
Exercise - 9: PPT Presentations/ Poster Presentation
Exercise - 10: Interviews Skills

TEXT BOOKS:

1. *Pathfinder: Communicative English for Undergraduate Students*, 1st Edition, Orient Black Swan, 2023.
2. *Empowering with Language*, 1st Edition, Cengage Learning India Pvt. Ltd., 2023.

REFERENCE BOOKS:

1. Dubey, Sham Ji & Co. *English for Engineers*. Vikas Publishers, 2021.
2. Lewis, Norman. *Word Power Made Easy- The Complete Handbook for*

Building aSuperior Vocabulary. Penguin and Goyal, 2026.
3. Wren P.C. and Martin H., *High School English Grammar and Composition*, S. Chand, Revised Edition, 2026.
4. J. Sethi & P.V. Dhamija. *A Course in Phonetics and Spoken English*, 2nd Edition, Phi learning, 2023.

ONLINE RESOURCES:

1. British Council – Learn English – https://learnenglish.britishcouncil.org
2. NPTEL – Communication Skills – https://nptel.ac.in/courses/109104101
3. Purdue OWL (Online Writing Lab) – https://owl.purdue.edu
4. BBC Learning English – https://www.bbc.co.uk/learningenglish
5. TED Talks – https://www.ted.com/talks (Listening & speaking practice)
6. Coursera English Communication Skills Specialization – https://www.coursera.org
7. EdX English for Professional and Academic Purposes – https://www.edx.org
8. ELSA Speak (App) – https://elsaspeak.com (AI pronunciation coach)

LAB SOFTWARE:

- Young India Films
- Walden InfoTech

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	–	–	1	–	–	2	3	–	–
CO2	3	2	1	1	2	–	–	2	3	–	–
CO3	2	3	1	1	1	–	–	3	3	–	–
CO4	2	3	2	2	2	–	–	2	3	–	–
CO5	2	2	1	1	2	–	–	3	3	–	–

Correlation Levels: 3- High; 2-Moderate; and 1- Low

I B. Tech I Semester – Common to EEE, CSE, CSD, CSM, CSE (AI&DE), CSE (AI&R) and CSBS
I B. Tech II Semester– Common to ECE, IT, CAI, CSC, CSO, AI&DS, AI, ECE (DT), CSE (CC&A) and CSE (QC)

L	T	P	C
3	0	2	4

26AHS05: APPLIED CHEMISTRY

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1 : Apply principles of quantum chemistry and molecular orbital theory to analyze molecular structure and bonding in technologically relevant materials.

CO2 : Derive and apply integrated rate equations to analyze zero-order, first-order and second-order chemical reactions.

CO3 : Evaluate the principles, performance and applications of batteries, supercapacitors and fuel cells for modern electronic and energy-storage systems.

CO4 : Analyze corrosion mechanisms in electronic components and recommend suitable protective techniques for PCBs, solder joints and interconnects.

CO5 : Analyze the synthesis, properties and applications of nanomaterials for nanoelectronics, sensors, memory and emerging technologies.

UNIT - I: QUANTUM CHEMISTRY 9 Hours

Fundamentals of Quantum Mechanics: Black body radiation, photoelectric effect, wave-particle duality, de Broglie wave equation, Heisenberg's uncertainty principle. Schrödinger wave equation; significance of Ψ and Ψ^2; quantum numbers, Particle in a one-dimensional box.

Molecular Orbital Theory: Bonding in homonuclear and heteronuclear diatomic molecules; energy level diagrams of O_2 and CO, π-molecular orbitals of butadiene and benzene; calculation of bond order.

AI Integration: Simulation of quantum systems using Python (NumPy, SciPy); visualization of molecular orbitals and energy levels using computational chemistry tools (Google Colab); introduction to quantum computing simulations using Qiskit (introductory level).

UNIT –II: CHEMICAL KINETICS 9 Hours

Fundamentals of Chemical Kinetics, Rate of a chemical reaction, Average and instantaneous rate, Rate law and rate constant, Order and molecularity of a reaction. Elementary and complex reactions, Difference between order and molecularity.

Factors affecting reaction rate: Concentration, Temperature, Pressure, Nature of reactants and Catalyst. Integrated Rate equations for Zero-order reactions, First-order reactions, Second-order reactions, Pseudo-first-order reactions, Examples: Hydrolysis of ester, Inversion of cane sugar, Arrhenius equation and Applications of chemical kinetics.

AI Integration: AI regression calculation and prediction of rate constant, AI assisted regression of ln k Vs 1/T

UNIT – III: ELECTROCHEMISTRY AND ENERGY STORAGE 9 Hours

Electrochemical cells: galvanic cells, electrode potential, Nernst equation and problems; potentiometry (redox reactions); conductometry (strong acid vs. strong base, weak acid vs. strong base), Primary batteries: Zinc-air battery, Sodium-air battery, Secondary batteries: chemistry of charging and discharging of lead-acid, nickel-cadmium, nickel-metal hydride, Lithium-ion batteries: construction, working, and applications of $LiCoO_2$, Supercapacitors: definition, classification, and applications. Fuel cells: construction, working, and applications of hydrogen fuel cells and direct methanol fuel cells (DMFC). Battery management systems: overcharge protection and thermal management.

AI Integration: AI models for battery health prediction and state-of-charge optimization using scikit-learn; data analysis for energy efficiency in data centres using Python (Pandas); ML-based prediction of electrochemical impedance spectra.

UNIT – IV: CHEMISTRY OF CORROSION AND PROTECTIVE COATINGS 9 Hours

Corrosion: definition, examples, types - chemical corrosion and electrochemical corrosion; Pilling-Bedworth rule, Galvanic corrosion; concentration cell corrosion; factors affecting corrosion. Deal-Grove model: thermal oxidation of silicon; passivation layers - silicon dioxide, silicon nitride, and aluminium oxide, Corrosion of printed circuit boards (PCBs): copper oxidation, tin whiskers; corrosion in solder joints and interconnects, Protective techniques: proper design, using pure metals, metal alloys, inhibitors; cathodic protection - sacrificial anodic protection, impressed current cathodic protection. Electroplating (nickel) and electroless plating (copper).

AI Integration: AI-based prediction of corrosion in electronic components, models for identifying corrosion types from microscope images.

UNIT -V: CHEMISTRY OF NANOMATERIALS 9 Hours

Introduction to nanochemistry: size-dependent properties, quantum confinement, Carbon nanomaterials: fullerenes, carbon nanotubes (CNTs), grapheme, Synthesis methods: chemical vapour deposition (CVD), sol-gel process, Quantum dots: synthesis, properties, applications in displays and quantum computing, Metallic nanoparticles: preparation and applications of gold, silver, and copper nanoparticles, Applications: nanoelectronics, interconnects, sensors (optical and biosensors), memory devices, Spintronic materials:

magnetic nanoparticles, topological insulators, Chemistry of quantum computing materials: superconducting materials (NbTi, Nb_3Sn); ion trap materials.

AI Integration: AI-based nanomaterial property prediction using machine learning (Python); simulation of nano-devices and quantum materials using open-source tools; generative AI for exploring nanomaterial design spaces.

LIST OF EXPERIMENTS:

1. 1.Estimation of Ferrous Ion (Fe^{2+}) using Potassium Permanganate ($KMnO_4$) by Redox Titration
2. Determination of Strength of Acid (H_2SO_4) in Battery Using Sodium Hydroxide (NaOH)
3. Determination of Rate of Corrosion by weight loss method.
4. Determination of Corrosion by Environment – pH, temp, moisture content.
5. Conductometric Titration of Strong Acid vs Strong Base (HCl vs NaOH)
6. Conductometric Titration of Weak Acid vs Strong Base (CH_3COOH vs NaOH)
7. Estimation of Iron (Fe^{2+}) Using Potassium Dichromate ($K_2Cr_2O_7$) by Potentiometry.
8. Determination of rate of Ester hydrolysis by pseudo first order reaction.
9. Determination of viscosity of PVA using Ostwald Viscometer.
10. Synthesis of ZnO Nano particles by precipitation method.
11. Estimation of Iron III in the given sample by colorimetry.
12. Determination of conductance of an electrolyte solution using conductivity meter.

TEXT BOOKS:

OF NANOMATERIALS | CHEMISTRY OF NANOMATERIALSHE

1. P. C. Jain & M. Jain. *Engineering Chemistry*, 17th Edition, Dhanpat Rai Publishing Company, 2025.
2. Shika Agarwal. *Engineering Chemistry* 5th Edition, Cambridge University Press, 2026.

REFERENCE BOOKS:

1. D. Linden & T. B. Reddy, (Eds.), *Handbook of Batteries*, 3rd Edition, McGraw-Hill., 2001.
2. C. N. R. Rao, A. Müller & A. K. Cheetham, (Eds.) *The Chemistry of Nanomaterials: Synthesis, Properties and Applications* (Vols. 1-2). Wiley-VCH, 2007.
3. C. P. Poole & F.J. Owens, *Introduction to Nanotechnology*. John Wiley & Sons, 2003.
4. B. K. Sharma & H. Gaur, *Engineering Chemistry*. Krishna Prakashan Media, 2019.
5. S. S. Dara & S. S. Umare, *Engineering Chemistry*, 15th Edition, S. Chand Publishing, 2018.

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	2	–	–	–	2	–	–
CO2	3	3	2	3	2	–	–	–	1	–	–
CO3	3	3	3	3	3	–	–	1	2	–	–
CO4	3	3	3	3	3	–	–	1	2	–	–
CO5	3	3	3	3	3	–	–	1	2	–	–

Correlation Levels: 3-High; 2-Moderate; and 1-Low

I B.Tech. I Sem- Common to All Branches

L	T	P	C
3	0	0	3

26AHS03: LINEAR ALGEBRA AND CALCULUS

Course Outcomes

After Successful completion of the course, the student will be able to:

CO1 : Apply matrix algebra techniques to analyze matrix properties and solve systems of linear equations using direct and iterative methods in engineering applications.

CO2 : Analyze Eigen value and eigenvector problems, apply diagonalization and Cayley- Hamilton theorem, reduce and classify quadratic forms using appropriate transformations.

CO3 : Apply mean value theorems, Taylor and Maclaurin series, curvature concepts to analyze and approximate engineering functions and curves.

CO4 : Analyze multivariable functions using partial and total derivatives, Jacobians and directional derivatives, solve unconstrained and constrained optimization problems.

CO5 : Evaluate double and triple integrals using appropriate coordinate transformations to determine areas, volumes, mass, centroids and moments in engineering applications.

UNIT – I: MATRIX ALGEBRA AND LINEAR SYSTEMS 9 Hours

Matrices, types of matrices, operations, transpose, inverse, symmetric and skew-symmetric matrices; determinant properties; rank of a matrix; echelon and normal forms; system of linear equations; consistency conditions; Gauss elimination and Gauss-Jordan methods; iterative methods: Jacobi and Gauss-Seidel; applications to engineering systems.

AI Integration: Python-based solution of large linear systems using NumPy; visualization of convergence in iterative methods; AI-assisted debugging of matrix algorithms; using ChatGPT/Copilot to explain row operations and rank reduction.

UNIT - II: EIGEN VALUES, EIGEN VECTORS, DIAGONALIZATION & QUADRATIC FORMS 9 Hours

Eigen values and eigenvectors; characteristic equation; Cayley-Hamilton theorem; diagonalization; similarity transformation; powers and inverse of matrices using Cayley-Hamilton theorem; quadratic forms; canonical form; reduction by orthogonal transformation; rank, index, signature; positive definite forms.

AI Integration: AI-assisted Eigen value computation and verification; Python/SciPy visualization of Eigen modes and quadratic surfaces; use of AI tools to explain diagonalization and matrix classification; numerical checking of Cayley-Hamilton theorem.

UNIT - III: SINGLE VARIABLE CALCULUS AND SERIES EXPANSIONS

9 Hours

Rolle's mean value theorem; Lagrange's mean value theorem; Cauchy's mean value theorem; Taylor's and Maclaurin's series with remainder; approximation of functions; curvature of plane curves; radius and centre of curvature; circle of curvature; applications to engineering approximations and curve design.

AI Integration: SymPy-based Taylor and Maclaurin expansion generation; plotting truncation error and convergence; AI-supported derivation checking for mean value theorems; curvature plot visualization and interpretation.

UNIT - IV: MULTI VARIABLE DIFFERENTIATION AND OPTIMIZATION

9 Hours

Functions of several variables: Continuity and Differentiability, Partial derivatives, total derivatives, chain rule, Directional derivative, Taylor's and Maclaurin's series expansion of functions of two variables. Jacobian, Functional dependence, maxima and minima of functions of two variables, method of Lagrange multipliers.

AI Integration: Gradient field visualization using Python; AI-assisted Jacobian computation and verification; constrained optimization using SciPy; use of AI tools to interpret saddle points, extrema, and optimization constraints.

UNIT-V: MULTIPLE INTEGRALS AND COORDINATE TRANSFORMATIONS

9 Hours

Double integrals; change of order of integration; triple integrals; change of variables; polar, cylindrical and spherical coordinates; area and volume computation; mass, centroid and moment calculations; engineering applications of multiple integration.

AI Integration: Numerical evaluation of double and triple integrals using SciPy; 2D and 3D region visualization; AI-assisted generation of integration code; interpreting Jacobian as geometric scaling factor in transformed coordinates.

TEXT BOOKS:

1. E. Keryszig, *Advanced Engineering Mathematics*, 11th Edition, International Adaptation, Hoboken, NJ, USA: Wiley, 2025.
2. B. S. Grewal, *Higher Engineering Mathematics*, 45th Edition, New Delhi, India: Khanna Publishers, 2024.
3. J. R. Hass, C. E. Heil, M. D. Weir, and P. Bogacki, *Thomas' Calculus*, 15th Edition, Pearson, 2022.
4. R. K. Jain and S. R. K. Iyengar, *Advanced Engineering Mathematics*, 5th Edition, Oxford, U.K.: Alpha Science International, 2021.

REFERENCE BOOKS:

1. G. Strang, *Introduction to Linear Algebra*, 6th Edition, Wellesley Cambridge Press, 2023, ISBN: 978-1-7331466-7-8.
2. D. C. Lay, S. R. Lay, and J. J. McDonald, *Linear Algebra and Its Applications*, 6th Edition. Pearson, 2021.
3. G. B. Arfken, H. J. Weber, and F. E. Harris, *Mathematical Methods for Physicists: A Comprehensive Guide*, 7th Edition, Academic Press (Elsevier), 2013, ISBN:978-0-12-384654-9.
4. D. G. Zill and W. S. Wright, *Advanced Engineering Mathematics*, 6th Edition. Jones & Bartlett Learning, 2018.

Free/Open-Source AI Tools /Software:

1. Google Colab
2. Python (NumPy, SymPy, SciPy, Matplotlib)
3. Scilab
4. Sage Math
5. GeoGebra

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	2	–	–	–	–	–	–
CO2	3	3	3	2	2	–	–	–	–	–	–
CO3	3	3	3	2	2	–	–	–	–	–	–
CO4	3	3	3	3	2	–	–	–	–	–	–
CO5	3	3	3	3	2	–	–	–	–	–	–

Correlation Levels: 3 - High; 2 - Moderate; and 1- Low

I B. Tech I Semester – Common to all Branches

L	T	P	C
3	0	2	4

26ACS01: COMPUTATIONAL THINKING AND PROBLEM SOLVING WITH PYTHON

Course Outcomes (COs)

After successful completion of the course, the student will be able to:

CO1 : Apply computational thinking techniques and Python programming fundamentals to design and implement basic programs using algorithms, flowcharts, pseudocode, variables, data types and operators.

CO2 : Apply decision-making and looping constructs to develop Python programs for solving conditional and iterative computational problems.

CO3 : Analyze problem requirements and apply Python string-handling techniques and core data structures such as lists, tuples and sets to develop data-processing solutions.

CO4 : Analyze computational problems and develop modular, reusable Python solutions using dictionaries, user-defined functions, recursion and lambda functions.

CO5 : Develop Python programs using modules, file handling and exception-handling mechanisms, apply basic object-oriented programming concepts to solve practical problems.

UNIT – I: COMPUTATIONAL THINKING AND PROGRAMMING BASICS

10 Hours

Computational Thinking: Characteristics, problem-solving strategies, steps in problem solving; algorithms - definition and properties; flowcharts - symbols and construction; pseudocode - writing and conversion; abstraction, decomposition, pattern recognition, algorithm efficiency basics.

Introduction to Python: Installation and execution environment; variables, identifiers, keywords; Data types, Type conversion; Input and Output statements; Expressions and Operators - Arithmetic, Relational, Logical, Assignment; Operator precedence.

AI Integration: Use Flowgorithm to design and simulate algorithms (e.g., largest of 3 numbers); Draw flowcharts using Lucidchart; Use Algorithm Visualizer to trace algorithm execution; Use ChatGPT to convert pseudocode to Python and verify output.

UNIT – II: DECISION MAKING AND LOOPING 8 Hours

Control Statements: Boolean expressions; if, if-else, if-elif-else, nested if; Conditional expressions (ternary operator). Looping statements: while loop, for loop, iteration techniques, nested loops, infinite loops; loop control - break, continue, pass; else with loops.

Problem Solving: Prime number check, Pattern programs using nested loops, menu-driven programs using control statements.

AI Integration: Use Python Tutor to visualize if-else and loop execution step by step; Use Thonny to debug loop-based programs, Use ChatGPT to generate test cases for control flow programs.

UNIT – III: STRINGS AND DATA STRUCTURES 9 Hours

Strings: Representation, indexing, slicing, operations, built-in functions and methods.

Lists: Creation, indexing and slicing, operations, functions, methods, nested lists.

Tuples: Creation, operations, packing and unpacking.

Sets: creation, set operations — union, intersection, difference; frozen sets.

AI Integration: Use NLTK and TextBlob to perform text analysis (sentiment, word frequency) on strings; Use Pandas and NumPy for list and array operations; Use ChatGPT to explain string slicing and list comprehensions with concrete examples.

UNIT – IV: FUNCTIONS AND PROBLEM SOLVING 9 Hours

Dictionaries: Creation, operations, methods; dictionary-based applications. Functions: Built-in and user-defined functions; function definition and calling; arguments - positional, keyword, default, variable-length; scope of variables (local and global).

Recursion: Recursive functions - factorial, Fibonacci; lambda functions (anonymous functions); Applications of functions in problem solving.

AI Integration: Use Jupyter Notebook for interactive function experiments; Use SymPy for symbolic mathematical expressions; Use PyCharm for modular program development.

UNIT – V: FILE HANDLING, EXCEPTION HANDLING AND OOP 9 Hours

Modules and Packages: Creating and importing modules; Standard library modules.

File handling: Opening, reading, writing, and closing files; file modes; working with text files; processing CSV and Excel files.

Exception handling: Types of errors (syntax, runtime, logical); try, except, finally blocks; raising exceptions. Introduction to OOP: classes, objects, attributes, methods, constructors, self keyword; basic applications of OOP in Python.

AI Integration: Use Pandas and OpenPyXL to read and write CSV/Excel files; Use Pytest for testing Exception handling; Use ChatGPT to explain OOP class design with real-world examples; Use Django for a simple mini web application as a capstone project.

LIST OF EXPERIMENTS:

Experiment 1: Select three daily-life problems; apply decomposition and abstraction; write algorithms, draw flowcharts, and convert them into pseudocode.

Experiment 2: Design flowcharts for largest of three numbers, student grade calculation, and electricity-bill generation; simulate execution and verify outputs.

Experiment 3: Install the Python environment; write programs using variables, data types, input/output, and operators; solve simple interest, area, and unit-conversion problems.

Experiment 4: Develop Python programs for largest among three numbers, grade calculation, leap-year checking, and a menu-driven calculator.

Experiment 5: Implement programs for prime-number generation, Armstrong numbers, factorial computation, and multiplication tables.

Experiment 6: Generate pyramid, diamond, and character patterns.

Experiment 7: Develop programs for string reversal, palindrome checking, frequency counting, and word-occurrence analysis.

Experiment 8: Perform sorting, searching, list merging, and matrix operations using nested lists.

Experiment 9: Implement tuple packing/unpacking, student-record storage, and set union, intersection, and difference operations.

Experiment 10: Implement positional, keyword, default, and variable-length arguments.

Experiment 11: Develop recursive solutions for Factorial, Fibonacci, GCD, and Tower of Hanoi.

Experiment 12: Implement a student database, dictionary operations, dictionary sorting, and lambda-based computations.

Experiment 13: Develop programs to create and read text files, count words, and process CSV data.

Experiment 14: Implement try-except, finally, and custom exceptions; apply exception handling to division operations, file access, and input validation.

Experiment 15: Develop a Python application using classes, objects, constructors, and methods.

TEXT BOOKS

1. P. Spronck, *Computational Thinking with Python*, Version 1.0.0, 2023.
2. R. Thareja, *Python Programming Using Problem-Solving Approach*, 3rd Edition. New Delhi, India: Oxford University Press, 2025.

REFERENCE BOOKS

1. A. B. Downey, *Think Python: How to Think Like a Computer Scientist,* 3rd Edition, O'Reilly Media, 2024.
2. E. Matthes, *Python Crash Course: A Hands-On, Project-Based Introduction to Programming*, 3rd Edition, No Starch Press, 2023.
3. J. VanderPlas, *Python Data Science Handbook: Essential Tools for Working with Data*, 2nd Edition, O'Reilly Media, 2023.

ONLINE RESOURCES

1. Algorithm & Flowchart: Flowgorithm – flowgorithm.org; Lucidchart lucidchart.com Algorithm Visualizer – algorithm-visualizer.org

2. Programming & Debugging: Python Tutor – pythontutor.com; Thonny – thonny.org Replit – replit.com
3. Data & Scientific: Pandas – pandas.pydata.org; NumPy – numpy.org; SymPy – sympy.org
4. NLP & Text: NLTK – nltk.org; TextBlob – textblob.readthedocs.io
5. Development: Jupyter Notebook – jupyter.org; PyCharm – jetbrains.com/pycharm OpenPyXL – openpyxl.readthedocs.io
6. NPTEL – Problem Solving through Programming in C / Python – nptel.ac.in

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	1	3	–	–	–	1	–	–	3	2
CO2	3	3	3	1	3	–	–	–	1	–	–	3	2
CO3	3	3	3	2	3	–	–	–	2	–	–	3	2
CO4	3	3	3	2	3	–	–	1	2	–	–	3	3
CO5	3	3	3	2	3	–	–	2	2	–	–	3	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B. Tech I Semester - Common to All Branches

L	T	P	C
1	0	4	3

26AAI01:AI TOOLS AND APPLICATIONS

Course Outcomes (COs)

After successful completion of the course, students will be able to:

CO1 : Explain the fundamental concepts of artificial intelligence and generative AI, classify and compare AI tools across text, image, audio, video and code applications.

CO2 : Apply prompt-engineering techniques and AI productivity tools to create, refine and evaluate documents, presentations and study materials using appropriate source-grounded workflows.

CO3 : Use AI-assisted research, note-taking and no-code data-analysis tools to retrieve, verify, organize, analyze and interpret information and datasets using appropriate evidence-verification practices.

CO4 : Create and evaluate AI-assisted visual, diagrammatic, audio and video content, identify and validate relevant AI applications for engineering disciplines using appropriate technical or authoritative sources.

CO5 : Evaluate AI applications for bias, privacy, intellectual property, ethical and governance risks, design and present an AI-assisted mini-project integrating appropriate tools and responsible-AI practices.

UNIT – I: UNDERSTANDING AI AND AI FOR DOCUMENTS AND PRESENTATIONS

15 Hours

AI Definition: AI problems, foundation of AI and AI brief history (Turing Test → ChatGPT → generative AI era); how LLMs work (training data, next-token prediction, temperature). AI ecosystem: ChatGPT, Claude, Gemini, Microsoft Copilot- strengths and differences.

AI for presentations: Gamma.app - full slide deck from a paragraph prompt; Microsoft Copilot in PowerPoint - slide generation, design suggestions, speaker notes; Canva AI - Magic Design, text-to-presentation, brand kits; Beautiful.ai -smart slide layouts; comparing AI-generated vs. manually created presentations.

AI for document writing: Microsoft Copilot in Word - draft, rewrite, summarize; Gemini in Google Docs - writing assistance and inline Q&A; Notion AI - meeting notes, project plans, knowledge base; Claude Projects - organizing documents and having a persistent AI assistant over a set of files.

AI writing assistance: Grammarly - grammar, tone, clarity, plagiarism; ChatGPT and Claude for lab reports, proposals, and letters; academic integrity - AI as an assistant not an author; disclosure norms; identifying AI-generated content using GPTZero.

AI Integration: Use Gamma.app to generate a presentation on an engineering topic in under

2 minutes; use Copilot in Word or Gemini in Docs to draft and improve a report; compare ChatGPT, Claude, Gemini, and Microsoft Copilot on the same writing task; use Grammarly to review your own writing.

UNIT – II: AI FOR RESEARCH, NOTE-TAKING AND SPREADSHEETS

15 Hours

NotebookLM (Google): uploading lecture notes, textbooks, PDFs, and research papers as sources; asking questions grounded in uploaded documents; generating study guides, timelines, briefing documents, and audio overviews (podcast-style summaries) from sources; source-grounded answers that cite the exact passage - eliminating hallucination risk.

AI for research: Perplexity AI (real-time web search with citations), Gemini Deep Research (multi-step research reports), Elicit and Consensus (academic paper search with evidence summaries); verifying AI-generated citations on Google Scholar; why AI fabricates references and how to detect it.

Prompt engineering: zero-shot, one-shot, few-shot, chain-of-thought, role-based prompting; iterative refinement; negative prompting; building effective prompts for summarization, comparison, explanation, and data extraction; using ChatGPT custom instructions and Claude Projects for persistent context.

AI for spreadsheets and data analysis: Microsoft Copilot in Excel - VLOOKUP, IF, COUNTIF, pivot tables from natural language; Gemini in Google Sheets - formula generation and chart creation; ChatGPT Advanced Data Analysis and Julius AI - uploading CSV files, computing statistics, generating charts, interpreting patterns - no coding required.

AI Integration: Upload your own lecture notes to NotebookLM and generate a study guide and audio overview; use Perplexity AI and Gemini Deep Research to research a topic; verify citations on Google Scholar; use Wolfram Alpha or Symbolab to solve and verify equations from your current semester subjects; use Copilot in Excel or ChatGPT ADA to analyse a real dataset.

UNIT – III: AI FOR VISUAL DESIGN, DIAGRAMS, AND MEDIA 15 Hours

AI for image generation: DALL-E 3 (via ChatGPT), Canva AI (text-to-image, Magic Design, background remover), Adobe Firefly, Ideogram - writing effective image prompts (style, subject, composition, colour); iterative prompt refinement for posters, diagrams, and mockups.

AI for diagrams and visual thinking: Napkin.ai - converting text or concepts into visual diagrams, charts, and infographics automatically; Whimsical AI - AI-generated mind maps, flowcharts, and wireframes from a description; Miro AI - collaborative whiteboard with AI summarization and diagramming.

AI for audio and video: ElevenLabs - text-to-speech and voice cloning for presentations and voiceovers; Suno and Udio - AI music generation for project videos; Runway and Pika — generating short video clips from text prompts; HeyGen - AI avatars for video presentations; current limitations and appropriate use cases.

AI content detection and critical evaluation: GPTZero, Originality.ai for AI-text detection; Google Lens and reverse image search for AI-generated images; fact-checking workflows - cross-referencing primary sources; deepfakes and synthetic media - recognition, risks, and responsible creation.

AI Integration: Use Napkin.ai to convert a paragraph of text into a diagram; use Whimsical AI to generate a mind map of a course topic; use DALL-E 3 or Canva AI to create an event poster through 3 prompt iterations; use ElevenLabs to create a voiceover; test GPTZero on 5 samples and analyse confidence scores.

UNIT– IV: AI ACROSS ENGINEERING DISCIPLINES 15 Hours

AI in Mechanical Engineering: generative design in CAD (Autodesk Fusion 360 - AI-optimized structures for weight and load); predictive maintenance using sensor data and ML; AI-assisted FEA simulation; manufacturing process optimization; digital twins.

AI in Civil and Electrical Engineering: structural health monitoring using ML; smart grid load forecasting and energy management; AI for circuit simulation and PCB layout optimization (Altium AI); VLSI design automation - floor planning and routing; power systems fault detection.

AI in Electronics and Communication: AI-assisted signal processing and filtering; speech recognition for embedded systems; computer vision for industrial quality inspection; AI in antenna design and spectrum management; MATLAB AI and Deep Learning Toolbox.

AI in Computer Science and allied branches: GitHub Copilot for code generation, explanation, and review; AI-powered testing (Testim, Selenium AI); AI in cybersecurity - anomaly detection and intrusion detection; no-code/low-code platforms (Bubble, Glide, Webflow AI); AI for data science (Jupyter AI, DataRobot).

AI Integration Each student identifies 5 real AI tools in their own engineering branch using Claude or ChatGPT and verifies each using official product pages or research publications; present findings in a 3-minute class presentation with a structured comparison table.

UNIT– V: RESPONSIBLE AI, ETHICS 15 Hours

AI bias and fairness: sources of bias (data, model, deployment); real documented cases - Amazon hiring algorithm (2018), COMPAS recidivism (2016), facial recognition disparities (NIST 2019); fairness metrics; misinformation, deepfakes, and synthetic media risks.

Privacy and intellectual property: what happens to data entered into public AI tools; PII risks; data retention policies of ChatGPT (OpenAI), Claude (Anthropic), and Gemini (Google); who owns AI-generated content; copyright status globally; academic plagiarism with AI; personal data safety protocol.

AI governance frameworks: EU AI Act risk tiers (unacceptable, high, limited, minimal); UNESCO AI Ethics Recommendation; India's NITI Aayog Responsible AI Guidelines; India Digital Personal Data Protection Act 2023 (DPDP Act); environmental cost of AI - energy and carbon footprint of LLM training and inference.

AI Integration: Read and compare data retention policies of ChatGPT, Claude, and Gemini; classify 5 daily-use AI systems using EU AI Act risk tiers; use GPTZero and Google Lens to detect AI-generated content; design and present the Capstone Mini-Project integrating tools from all five units.

LIST OF EXPERIMENTS

Sl. No.	Unit	Title of the Experiment
1	Unit I	ChatGPT & Claude - Understanding LLMs, Temperature and Hallucination; Unit I: Understanding AI and AI for Documents
2		Gamma.app - AI-Generated Presentation from a Paragraph Prompt; Unit I: AI for Presentations
3		Microsoft Copilot in Word - Draft, Rewrite, and Summarize; Unit I: AI for Document Writing
4		Grammarly - Grammar, Tone, Clarity, and Plagiarism Detection; Unit I: AI Writing Assistance
5		GPTZero - AI Text Detection: Test 5 Samples and Analyze Scores; Unit I: Academic Integrity & AI Detection
6		Google Gemini - Multimodal Prompting with Image and Text Input
7	Unit II	NotebookLM - Upload Lecture Notes and Generate Study Guide + Audio Overview; Unit II: NotebookLM
8		Perplexity AI - Real-Time Research with Verified Citations; Unit II: AI for Research
9		Elicit & Consensus - Academic Paper Search and Evidence Synthesis; Unit II: AI for Research
10		Prompt Engineering - Zero-Shot to Chain-of-Thought on Engineering Tasks; Unit II: Prompt Engineering
11		Copilot in Excel / Julius AI - Natural Language Data Analysis of a CSV Dataset; Unit II: AI for Spreadsheets
12		Zotero + SciSpace - Reference Management and AI Paper Q&A
13	Unit III	DALL-E 3 / Canva AI - Event Poster Through 3 Iterative Prompt Refinements; Unit III: AI for Image Generation
14		Napkin.ai & Whimsical AI - Concept Diagram and Course Mind Map; Unit III: AI for Diagrams
15		Miro AI - Collaborative System Architecture Brainstorm; Unit III: AI for Diagrams
16		Eleven Labs - Create a 60-Second Technical Project Voiceover; Unit III: AI for Audio
17		Google Lens & GPTZero - Detect AI-Generated Text and Images; Unit III: AI Content Detection
18		Text-to-Video - Create a 30-Second Technical Explainer with Canva AI / Runway
19	Unit IV	GitHub Copilot - Code Generation, Explanation and Review; Unit IV: AI in Computer Science
20		ChatGPT ADA - Full Data Science Pipeline on a Real Dataset; Unit IV: AI for Data Science
21		Bubble.io or Glide - Build a No-Code AI-Powered App Prototype; Unit IV: AI in CS - No-Code Platforms

22		Wolfram Alpha & Symbolab - Solve and Verify Engineering Equations; Unit IV: AI Math Verification
23		Branch AI Tool Research - Identify and Verify 5 Real Branch - Specific AI Tools; Unit IV: AI Tools Integration
24		Teachable Machine - Train and Deploy a No-Code Image Classifier
25	Unit V	AI Bias Audit - Analyze Compas Fairness Metrics; Unit V: AI Bias and Fairness
26		Privacy Audit - Compare Data Retention Policies of ChatGPT, Claude, Gemini; Unit V: Privacy and AI
27		EU AI Act Classification - Classify 5 Daily-Use AI Systems; Unit V: AI Governance
28		AI Hallucination & Fact-Checking Audit - A Responsible Verification Workflow
29		Capstone Deliverable - Integrated Multi-Tool Engineering Solution; Unit V: Capstone Project
30		Capstone Presentation - 5-Minute Project Demo with Responsible AI Disclosure; Unit V: Capstone Project

TEXT BOOKS:

1. S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th Edition. Pearson, 2022.
2. H. Motwani, *Generative AI 360°: Practical Guide to ChatGPT, Midjourney & AI Tools to Boost Productivity & Creativity: For Professionals,* 1st Edition ZebraLearn, 2025

REFERENCE BOOKS:

1. E. Mollick, *Co-Intelligence: Living and Working with AI*. Penguin, 1st Edition, 2024
2. Chip Huyen, *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*, 1st Edition, O'Reilly Media, 2022.
3. Reid Blackman, *Ethical Machines: Your Concise Guide to Totally Unbiased, Transparent, and Respectful AI*, Harvard Business Review Press, 2022
4. F. Chollet, *Deep Learning with Python*, 2nd Edition, Manning Publications, 2021

Course articulation matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1	–	–	3	–	–	–	1	–	–	3	2
CO2	3	2	2	1	3	–	–	–	3	–	–	2	3
CO3	3	3	1	3	3	–	–	–	2	–	–	3	3
CO4	3	2	3	2	3	–	–	1	2	–	–	3	3
CO5	3	2	3	2	3	–	–	2	3	–	–	3	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B. Tech I Semester - Common to CSE and Allied Branches
I B. Tech II Semester - Common to EEE, ECE & ECE (DT)

L	T	P	C
0	0	3	1.5

26AIT01: IT WORKSHOP

Course Outcomes (COs)

After successful completion of the course, students will be able to:

CO1 : Identify, configure and troubleshoot basic computer hardware components and peripheral devices.

CO2 : Install and configure Windows and Linux operating systems and perform basic file, user, process and package management using graphical and command-line tools.

CO3 : Configure basic TCP/IP network settings and troubleshoot common connectivity problems using appropriate network utilities.

CO4 : Apply essential cyber hygiene practices, including password management, firewall configuration, malware detection, safe browsing and multi-factor authentication.

CO5 : Create, format and manage professional documents, spreadsheets and presentations using office productivity tools.

CO6 : Use cloud-based collaboration and AI-assisted productivity tools responsibly to support academic, professional and team-based tasks.

MODULE I – PC HARDWARE AND TROUBLESHOOTING

EXPERIMENT-1: Identification of computer hardware components, peripherals, motherboard, processor, RAM, ROM, storage devices, expansion cards, SMPS, BIOS, CMOS, and I/O ports.

EXPERIMENT-2: Disassembly and assembly of desktop computer systems. Identification of internal components and cable connections.

EXPERIMENT-3: Installation and configuration of peripheral devices including printers, scanners, webcams, speakers, projectors, and external storage devices.

EXPERIMENT-4: BIOS configuration, boot sequence management, partitioning of storage devices, and preventive maintenance of computer systems.

EXPERIMENT-5: Hardware troubleshooting techniques including memory testing, hard disk diagnostics and power supply testing, and peripheral fault identification.

MODULE II – OPERATING SYSTEMS

EXPERIMENT-6: Installation of Microsoft Windows Operating System with device driver configuration.

EXPERIMENT-7: Installation of Linux Operating System using virtual machine environment.

EXPERIMENT-8: Practice Linux command-line interface:

- File management
- Directory operations
- File permissions
- User management
- Package installation
- Process management

MODULE III – NETWORKING AND INTERNET

EXPERIMENT-9: Configure TCP/IP settings, IP addressing, DNS, Gateway configuration, and network troubleshooting using command-line utilities.

EXPERIMENT-10: Internet services including:

- Email configuration
- Usage Search engines
- Cloud storage management
- File sharing techniques
- Online collaboration

MODULE IV- CYBERSECURITY

EXPERIMENT- 11: Cyber Hygiene Practices

- Antivirus installation
- Firewall configuration
- Password management
- Malware detection
- Safe browsing
- Multi-factor authentication

MODULE V – OFFICE PRODUCTIVITY TOOLS MS WORD / LIBREOFFICE WRITER

EXPERIMENT-12: Create professional documents and institutional documents using:

- Page formatting, styles, headers & footers
- Tables, images and references
- Mail Merge and Track Changes
- Resume preparation
- Technical report preparation
- Project certificate and newsletter
- Circular and brochure preparation
- Professional document formatting and review

MS EXCEL / LIBREOFFICE CALC

EXPERIMENT-13: Perform spreadsheet operations including:

- Formulae and functions
- Charts and conditional formatting
- Pivot Tables
- Data Validation
- Sorting & Filtering

- VLOOKUP/XLOOKUP
- Financial calculations
- Data analysis and visualization
- Spreadsheet-based problem solving

TEXT BOOKS:

1. J. Andrews, J. Shelton, S. Schmelz, and T. Jaqua Jr., *CompTIA A+ Guide to IT Technical Support*, 12th ed. Boston, MA, USA: Cengage, 2026 - Best primary text for hardware identification, assembly, troubleshooting, Windows/Linux, networking, security, cloud, virtualization and AI-assisted IT support. Cengage currently lists the 12th edition and specifically includes Windows 10/11, Linux, hardware troubleshooting, cloud, virtualization, security and AI.
2. A. S. Tanenbaum and H. Bos, *Modern Operating Systems*, 5th ed. Harlow, U.K.: Pearson, 2024 - Suitable for Module II, particularly processes, file systems, I/O, memory, Windows 11 and modern OS concepts.
3. R. Nordell, K. Stewart, A. Easton, P. R. Graves, and A. Mayhall, *Microsoft Office 365: In Practice, 2024 Edition*. New York, NY, USA: McGraw Hill, 2024 - Recommended for Module V, particularly Word, Excel, PowerPoint, collaboration and professional productivity. McGraw Hill also indicates a 2025 release, so this title can be updated to the current release if your institution prefers the newest version.

REFERENCE BOOKS:

1. E. Bott, *Windows 11 Inside Out*. Redmond, WA, USA: Microsoft Press, 2023 - Useful for Windows installation, configuration, administration and troubleshooting.
2. C. J. Pfleeger, S. L. Pfleeger, and L. Coles-Kemp, *Security in Computing*, 6th ed. Boston, MA, USA: Addison-Wesley Professional, 2023 - Additional reference for cybersecurity, authentication, secure systems, network security and modern security threats.
3. R. R. Panko and R. J. Boyle, *Corporate Cybersecurity*, 6th ed. Boston, MA, USA: Pearson, 2024 - Useful supplementary reference for firewalls, access control, host hardening, data protection and incident response.

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	2	3	-	-	2	2	-	-	3	2
CO2	3	2	2	3	3	-	-	2	2	-	-	3	2
CO3	3	3	2	3	3	-	-	2	2	-	-	2	3
CO4	3	3	3	3	3	-	-	2	2	-	-	2	3
CO5	3	3	3	3	3	-	-	2	3	-	-	2	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B.Tech. I Sem- Common to All Branches

L	T	P	C
0	0	1	0.5

26AHS08 - HEALTH AND WELLNESS, YOGA AND SPORTS

Course Outcomes (COs)

After the completion of the course the student will be able to

- CO1 : Explain and apply basic principles of health, nutrition, balanced diet and physical fitness in maintaining personal wellness.
- CO2 : Demonstrate appropriate yoga practices, including Asanas, Pranayama, Kriya, Mudra, Bandha and Dhyana for physical and mental well-being.
- CO3 : Demonstrate physical fitness and sports skills through participation in selected individual and team sporting activities and fitness exercises.
- CO4 : Demonstrate discipline, sportsmanship, emotional balance and a positive attitude while participating in sports, yoga and wellness activities.
- CO5 : Demonstrate teamwork and leadership skills through participation in sports, health-awareness and wellness activities.
- CO6 : Adopt healthy, responsible and sustainable lifestyle practices that contribute to personal and social well-being.

UNIT – I:

Concept of health and fitness, Nutrition and Balanced diet, basic concept of immunity Relationship between diet and fitness, Globalization and its impact on health, Body Mass Index (BMI) of all age groups.

ACTIVITIES:

i) Organizing health awareness programmes in community
ii) Preparation of health profile
iii) Preparation of chart for balance diet for all age groups

UNIT – II:

Concept of yoga, need for and importance of yoga, origin and history of yoga in Indian context, classification of yoga, Physiological effects of Asanas- Pranayama and meditation, stress management and yoga, Mental health and yoga practice.

ACTIVITIES:

Yoga practices – Asana, Kriya, Mudra, Bandha, Dhyana, Surya Namaskar

UNIT – III:

Concept of Sports and fitness, importance, fitness components, history of sports, Ancient and Modern Olympics, Asian games and Commonwealth games.

ACTIVITIES:

i) Participation in one major game and one individual sport viz., Athletics, Volleyball, Basketball, Handball, Football, Badminton, Kabaddi, Kho-kho, Table tennis, Cricket etc.
ii) Practicing general and specific warm up, aerobics
iii) Practicing cardiorespiratory fitness, treadmill, run test, 9 min walk, skipping and running.

GENERAL GUIDELINES:

1 . Institutes must assign slots in the Timetable for the activities of Health/Sports/Yoga.
2. Institutes must provide field/facility and offer the minimum of five choices of as many as Games/Sports.
3. Institutes are required to provide sports instructor / yoga teacher to mentor the students.

EVALUATION GUIDELINES:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit. Each activity shall be evaluated by the concerned teacher for 15 marks, totalling to 90 marks.
- A student shall be evaluated by the concerned teacher for 10 marks by conducting viva voce on the subject.

TEXT BOOKS:

1. Ministry of Youth Affairs and Sports, Government of India, NSS Training Manual. New Delhi, India: MYAS Publication, 2022.
2. Directorate General NCC, National Cadet Corps - Training Syllabus & Manual, New Delhi, India: DG NCC, 2021.

REFERENCE BOOKS:

1. S. K. Lal, Community Development & Social Service. New Delhi, India: Regal Publications, 2019.
2. S. Modh, Disaster Management – A Holistic Approach. New Delhi, India: Macmillan India, 2020.
3. S. McCurley and R. Lynch, Volunteer Management. Energize Publications, 2018.
4. R. Baden-Powell, Scouting for Boys, Centenary ed. Oxford, U.K.: Oxford University Press, 2008.

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	–	–	–	–	2	1	1	–	–	3
CO2	1	–	–	–	–	2	–	1	1	–	3
CO3	1	–	–	–	–	1	–	2	3	1	2
CO4	1	–	–	–	–	2	–	3	3	2	2
CO5	1	–	–	–	–	2	–	2	3	3	2
CO6	1	–	–	–	–	3	2	2	2	1	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B. Tech- I Semester – Common to ECE, ECE (DT), IT, CAI, CSC, CSO, AI, AI & DS, CSE (CC) and CSE (QT) branches.
I B. Tech- II Semester – Common to EEE, CSE, CSD, CSM, CSE (AI & DE), CSE (AI&R) and CSBS branches.

L	T	P	C
3	0	2	4

26AHS06: APPLIED PHYSICS

Course Outcomes (COs):

After successful completion of the course, a student will be able to:

CO1 : Apply the principles of wave optics to analyze interference, diffraction and polarization phenomena, determine relevant optical parameters through experiments, relate them to engineering applications.

CO2 : Demonstrate the principles of laser physics and optical fibres through appropriate experiments, determine laser and fibre-optic parameters, evaluate their applications in sensing and communication systems.

CO3 : Analyze the electrical and optical characteristics of semiconductor and optoelectronic devices, determine relevant device parameters through experiments, interpret their behavior for engineering applications.

CO4 : Analyze the dielectric and magnetic properties of engineering materials, determine relevant parameters through experimental investigation, relate polarization, dielectric loss and magnetic hysteresis characteristics to engineering applications.

CO5 : Analyze the fundamental properties and critical parameters of superconductors and nanomaterials, interpret experimental or simulated characterization results, evaluate their potential engineering applications.

UNIT-I: WAVE OPTICS 9 Hours

Interference: Interference of light - Conditions for sustained interference - Interference due to thin films (division of amplitude) - Newton's rings - Determination of wavelength and refractive index of a liquid.

Diffraction: Diffraction - types of diffraction - Fraunhofer diffraction - Single slit and double slit diffraction (qualitative) - diffraction grating - dispersive power and resolving power of grating.

Polarization: Polarization of light - production of Polarized light - reflection, refraction and double refraction - Nicol's prism-Half wave and Quarter wave plate -Applications of wave optics (CD/DVD data storage, Barcode scanners) in engineering and technology.

AI Integration: AI-based image analysis - Automatic identification of bright and dark fringes, AI pattern classification - Estimation of slit width and wavelength, AI-assisted

polarization management for quantum key distribution. AI-integrated holographic interferometer for surface flatness analysis.

UNIT-II: LASERS AND OPTICAL FIBRES 9 Hours

Lasers: Principle of lasing action – Characteristics - Einstein theory of stimulated emission - Population inversion-Pumping Schemes-Three and four level lasers-Components of laser - Construction and working of He-Ne Laser and Nd-YAG Laser -Applications of lasers in engineering and technology.

Optical Fibres**:** Introduction to fibre optics - Total Internal Reflection - critical angle - construction of optical fibers - Acceptance angle, Acceptance cone -Numerical Aperture - Classification of fibers based on Refractive index profile and modes of propagation- Attenuation (Losses) in optical fibers- Applications of an optical fiber in sensing communication**.**

AI Integration: AI-powered signal analysis for distributed fiber optic sensing. Python simulation of laser propagation and fiber optic signal attenuation.

UNIT-III: SEMICONDUCTORS AND OPTOELECTRONIC DEVICES 10 Hours

Semiconductors**:** Introduction - Intrinsic semiconductors - density of electron and density of holes -carrier concentration - fermi energy - electrical conductivity of intrinsic semiconductors - Extrinsic semiconductors - Density of charge carriers in n-type and p-type (qualitative) - Drift and Diffusion currents - Hall effect, Hall coefficient - Applications of Hall effect.

Optoelectronic devices**:** P-N junction-characteristics of P-N junction, Optoelectronic devices: Light emitting diode, Semiconductor diode laser and Solar cell.

AI Integration: AI-Assisted Hall Effect Analysis for Semiconductor Characterization. LEDS-AI-based optimization- Brightness control, energy efficiency and lifetime prediction, Intelligent feedback and control, Semiconductor laser-Intelligent feedback and control-Wavelength stabilization, power optimization and fault detection.

UNIT-IV: DIELECTRICS AND MAGNETISM 9 Hours

Dielectrics**:** Introduction-Electric polarization-Dielectric polarizability, susceptibility and dielectric constant-Types of polarizations (qualitative)-Frequency dependence of polarization-Lorentz (Internal) field-Dielectric loss-Piezoelectricity and ferroelectrics with applications.

Magnetism**:** Introduction-Magnetic dipole moment, magnetization, magnetic susceptibility and permeability-Origin of magnetic moment-Classification of magnetic materials-Hysteresis loop-Soft and hard magnetic materials- properties and applications.

AI Integration: AI Integration - Data-driven analysis - Prediction of polarization behaviour with frequency and temperature. AI-Assisted Magnetic Data Storage and MRAM Optimization.

UNIT-V: SUPERCONDUCTORS AND NANOMATERIALS 8 Hours

Superconductors**:** Introduction – Properties of Superconductors - Critical parameters of Superconductors - Meissener effect - Penetration depth - Types of Superconductors - BCS

Theory - Josephson effect (AC & DC) - High T_C Superconductors – Applications of Superconductors.

Nanomaterials**:** Introduction – significance of nanoscale - classification of nano materials - Synthesis of nanomaterials by Sol Gel, Hydrothermal and High Energy Balling method - Properties of nanomaterials - Applications of nanomaterials.

AI Integration: AI-Assisted Stability Control in Superconducting Magnetic Levitation Systems. AI Integration in Nanomaterials- ML-based property prediction- Designing nanomaterials with targeted properties.

LIST OF EXPERIMENTS:

1. Determination of radius of curvature of a plano convex lens using Newton's rings method.
2. Determination of the dispersive power of the prism in minimum deviation method.
3. Determination of Brewster's angle and refractive index of a given material
4. Determination of numerical aperture and acceptance angle of a given optical fiber.
5. Estimation of the particle size using a diode laser.
6. Determination of wavelength of a given He- Ne LASER.
7. Determination of Hall coefficient and number of charge carriers of a given semiconductor using Hall Effect experiment.
8. Determine the energy gap of a semiconductor using four probe method.
9. Verify the I-V characteristics of p-n junction diode and calculate its resistance under forward and reverse bias.
10. Estimation of threshold voltages of a given LED's.
11. Plotting the B–H hysteresis curve of a ferromagnetic material and determination of retentivity, coercivity and hysteresis loss.
12. Estimation of magnetic field along the axis of a circular coil carrying current.
13. Simulation of the superconducting transition: Plot resistance versus temperature and determine the critical temperature, followed by visualization of the Meissner effect (Virtual Lab).
14. I-V characteristics of a Josephson junction and its magnetic field dependence. (Virtual Lab)
15. Simulation of nanoparticle characterization- particle size dependent on electrical and optical properties (Virtual Lab).

TEXT BOOKS:

1. M.N. Avadhanulu, P.G. Kshirsagar & T.V.S Arun Murthy, *A Text book of Engineering Physics*, 11th Edition, S. Chand Publishing, 2026.
2. C.L. Arora, Practical Physics, 1st Edition, S. Chand publishing, New Delhi, 2023.
3. S. Balasubramanian, S. Ranganathan and M.N. Srinivasan, *A Text book of Practical Physics*, 2nd Edition, Sultan Chand Publishers, New Delhi, 2017.
4. P. K. Palaniswamy, *Engineering Physics*, 4th Edition, Scitech Publications India Private Limited, 2016.

REFERENCE BOOKS:

1. V. Rajendran, *Engineering Physics*, 2nd Edition, Mc Graw Hill, 2026.
2. K. Thyagarajan, *Engineering Physics*, McGraw-Hill Education (India) Pvt. Ltd, 2019.
3. R.K. Shukla and Anchal Srivastava, *Experimental Physics*, 1st Edition, New Age international publishers, 2017.
4. R. Jayaraman, V. Umadevi, S, Maruthamuthu, D. Sarvanakumar, *Engineering physics Laboratory Manual*,1st Edition, Pearson, 2013.

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	3	3	-	-	2	2	-	-
CO2	3	3	3	3	3	-	-	2	3	-	-
CO3	3	3	3	3	3	-	-	2	2	-	-
CO4	3	3	3	3	3	-	-	2	2	-	-
CO5	3	3	3	3	3	-	-	2	3	-	-

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B. Tech II Semester – Common to all Branches

L	T	P	C
3	0	2	4

26ACS02: DATA STRUCTURES USING PYTHON

Course Outcomes (COs)

After successful completion of the course, the student will be able to:

CO1 : Analyze abstract data types and data structures using time and space complexity measures, implement and compare basic searching and sorting algorithms.

CO2 : Implement stacks and queues using appropriate representations and apply them to expression evaluation, parenthesis matching, scheduling and other computational problems.

CO3 : Implement and manipulate singly, doubly and circular linked lists, apply them to practical problems such as polynomial processing, browser history and symbol tables.

CO4 : Analyze and implement binary trees, binary search trees and heaps, apply tree traversal, searching and heap operations, including heap sort, to computational problems.

CO5 : Represent and traverse graphs using adjacency matrices/lists, BFS and DFS, implement hash tables using appropriate collision-resolution techniques.

UNIT – I: INTRODUCTION, ALGORITHM ANALYSIS AND LINEAR STRUCTURES 10 Hours

Introduction to data structures, Abstract Data Types (ADTs); Algorithm analysis - time and space complexity; Big-O, Big-Omega, Big-Theta notations.

Lists in Python: operations on lists; static vs. dynamic memory allocation.

Searching techniques: Linear Search, Binary Search, Interpolation Search -analysis and comparison.

Sorting techniques: Bubble Sort, Selection Sort, Insertion Sort - algorithms, trace, and complexity analysis.

AI Integration: Use VisuAlgo and Algorithm Visualizer to animate sorting and searching; Use ChatGPT to explain Big-O complexity with concrete examples; Use Python Tutor to trace memory representation of data structures.

UNIT – II: STACKS AND QUEUES 8 Hours

Stacks: LIFO principle, push, pop, peek; Implementation using Python lists; applications - Expression evaluation (infix to postfix, postfix evaluation), parenthesis checking.

Queues: FIFO principle, enqueue, dequeue; types - Circular Queue, Priority Queue, Deque; implementation using Python lists and linked structures. Applications of queues: BFS simulation, scheduling, real-world analogies.

AI Integration: Use VisuAlgo to animate stack and queue operations; Use ChatGPT to generate edge cases (overflow, underflow, circular behaviour); Use GitHub Copilot to implement expression evaluation.

UNIT – III: LINKED LISTS 10 Hours

Singly Linked List: Node structure, insertion at head/tail/position, deletion by value/position, traversal, reversal.

Doubly Linked List: Forward and backward traversal, insertion, deletion; Circular Linked List: traversal and operations.

Applications: Polynomial representation and addition, browser history simulation (doubly linked list), symbol table implementation.

AI Integration: Use Python Tutor to trace pointer operations and node memory; Use VisuAlgo to visualize insertion and deletion step by step.

UNIT - IV: TREES AND HEAPS 9 Hours

Trees: Terminology, properties; Binary Trees - Representation, traversals (Inorder, Preorder, Postorder, level order).

Binary Search Trees (BST): Insertion, deletion, search; applications - autocomplete, range queries, dictionary lookup.

Heaps: Min-Heap and Max-Heap, array representation, heapify, Build-Heap; Heap Sort - algorithm and complexity.

AI Integration: Use VisuAlgo and Algorithm Visualizer to animate tree traversals and BST operations; Use GitHub Copilot to implement heap operations in Python.

UNIT – V: GRAPHS AND HASHING 9 Hours

Graphs: Terminology, types (directed, undirected, weighted); representation - adjacency matrix and adjacency list.

Graph traversals: BFS (queue-based) and DFS (stack/recursion-based); Applications - shortest path, cycle detection, social networks.

Hashing: Hash functions, properties of a good hash function; collision resolution - chaining (separate chaining), open addressing (linear probing, quadratic probing, double hashing); load factor.

AI Integration: Use NetworkX in Python and Gephi to build and traverse graphs; Use VisuAlgo to animate BFS and DFS; Use GitHub Copilot to implement hash tables with collision resolution.

LIST OF EXPERIMENTS

Experiment-1: Implement insertion, deletion, searching, and slicing; compare static and dynamic memory allocation using Python lists.

Experiment-2: Implement Linear Search, Binary Search, and Interpolation Search; compare

execution tie and number of comparisons.

Experiment-3: Implement Bubble Sort, Selection Sort, and Insertion Sort; visualize execution and compare performance on different input datasets.

Experiment-4: Implement stack operations using Python lists and solve parenthesis-matching problems.

Experiment-5: Implement infix-to-postfix conversion and postfix evaluation.

Experiment-6: Implement Queue, Circular Queue, Priority Queue, and Deque.

Experiment-7: Simulate BFS traversal and scheduling applications using queue structures.

Experiment-8: Implement insertion, deletion, traversal, and reversal operations for Single Linked List.

Experiment-9: Implement forward and backward traversal, insertion, and deletion for Double and Circular Linked List.

Experiment-10: Develop polynomial addition using linked lists and simulate browser history navigation.

Experiment-11: Implement Inorder, Preorder, Postorder, and level-order traversals for bianry Tree.

Experiment-12: Implement BST insertion, deletion, and search; implement Min Heap, Max Heap, and Heap Sort.

Experiment-13: Represent graphs using adjacency matrix and adjacency list; implement BFS and DFS.

TEXT BOOKS

1. S. K. Vasudevan, A. S. Nagarajan, and K. Nanmaran, *Data Structures Using Python,* 1st Edition, New Delhi, India: Wiley India, 2021.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser , *Data Structures and Algorithms in Python*, 1st Edition. New Delhi, India: Wiley India, 2021.

REFERENCE BOOKS

1. R. Thareja, *Python Programming: Using Problem-Solving Approach*, 3rd Edition, Oxford University Press, 2025.
2. E. Matthes, *Python Crash Course: A Hands-On, Project-Based Introduction to Programming,* 3rd Edition San Francisco, CA, USA: No Starch Press, 2022.
3. B. N. Miller, D. L. Ranum, and R. Yasinovskyy, *Problem Solving with Algorithms and Data Structures Using Python,* 3rd Edition. Franklin, Beedle & Associates, 2023.
4. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 4th Edition, MIT Press, 2022.

ONLINE RESOURCES

1. NPTEL – Data Structures and Algorithms – Nptel.ac.in
2. VisuAlgo – visualgo.net
3. Python Tutor – pythontutor.com
4. NetworkX – networkx.org
5. GeeksforGeeks – geeksforgeeks.org
6. Algorithm Visualizer – algorithm-visualizer.org

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	2	3	–	–	–	1	–	–	3	2
CO2	3	3	3	2	3	–	–	1	1	–	–	3	3
CO3	3	3	3	2	3	–	–	1	1	–	–	3	3
CO4	3	3	3	2	3	–	–	1	1	–	–	3	3
CO5	3	3	3	2	3	–	–	1	2	–	–	3	3

Correlation Levels: 3-High; 2-Moderate; and1-Low

I B. Tech II Semester- Common to All Branches

L	T	P	C
3	0	0	3

26AHS07: DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

Course Outcomes

After Successful completion of the course, the student will be able to:

CO1 : Solve first-order ordinary differential equations using appropriate analytical methods and apply them to model engineering processes such as cooling, growth / decay and electrical circuits.

CO2 : Analyze higher-order linear ordinary differential equations using appropriate solution methods and apply them to model dynamic systems such as electrical circuits, oscillators and mass-spring-dashpot systems.

CO3 : Formulate and solve first- and second-order partial differential equations using appropriate analytical methods, classify and interpret PDE models such as the heat, wave and Laplace equations.

CO4 : Compute and analyze the gradient, directional derivative, divergence and curl of Scalar and vector fields, apply vector identities and field concepts to engineering problems.

CO5 : Evaluate line and surface integrals and apply Green's, Stokes' and Divergence Theorems to compute work, circulation, flux, area and volume in engineering applications.

UNIT-I: FIRST-ORDER ORDINARY DIFFERENTIAL EQUATIONS 9 Hours

Classification of ODEs by order, degree, and linearity; linear first-order ODEs and integrating factor method; Bernoulli's equation; exact equations and integrating factors; orthogonal trajectories; Newton's law of cooling and natural growth/decay models; electrical circuit models (RC and RL circuits); engineering applications of first-order ODEs.

AI Integration: Python SymPyd solve for symbolic ODE solution and verification; direction field (slope field) visualization using Matplotlib; AI-assisted debugging of integrating factor derivations; using ChatGPT/Copilot to explain solution families and physical interpretations.

UNIT-II: HIGHER-ORDER LINEAR DIFFERENTIAL EQUATIONS 9 Hours

Superposition principle; Wronskian and linear independence; complementary function for constant-coefficient ODEs (distinct real, repeated, and complex roots); particular integral by operator method; variation of parameters; simultaneous linear ODEs; Cauchy's (Euler) equation; Legendre's equation; applications to LCR circuits, simple harmonic motion,

damped and forced oscillations; mass-spring-dashpot systems**.**

AI Integration: Python SciPy signal. 1sim for simulation of under damped, critically damped, and over damped responses; resonance amplitude curve generation; AI-assisted Wronskian computation and verification using SymPy; using AI tools to explain transient vs. steady-state response; resonance and Q-factor interpretation.

UNIT - III: PARTIAL DIFFERENTIAL EQUATIONS 9 Hours

Introduction and classification of PDEs; formation by elimination of arbitrary constants and functions; first-order linear PDEs and Lagrange's method; homogeneous linear PDEs with constant coefficients (CF and PI); standard forms of nonlinear first-order PDEs; second-order PDE applications - wave equation, heat equation, and Laplace's equation; classification as hyperbolic, parabolic, and elliptic.

AI Integration: Python SymPy for PDE formation and symbolic verification; 3D surface plot visualization of PDE solution families using Matplotlib; SciPy finite-difference simulation of the 1D heat equation with animation; AI-assisted PDE classification using the discriminant; prompting AI to classify mixed-derivative PDEs and student verification of AI reasoning.

UNIT - IV: VECTOR DIFFERENTIATION 9 Hours

Scalar and vector fields; del (∇) operator; gradient and directional derivative; unit - normal to surfaces; angle between surfaces; divergence and physical interpretation; curl and physical interpretation; vector identities; conservative fields and scalar potentials; Laplacian; engineering applications in electromagnetic and fluid mechanics contexts.

AI Integration: Python NumPy and Matplotlib for 2D and 3D gradient, divergence, and curl field visualization (quiver plots, level-surface overlays); SymPy for symbolic verification of vector identities; using AI tools to interpret Maxwell’s equations in terms of divergence and curl operators; AI-assisted explanation of irrotational and solenoidal field classification.

UNIT - V: VECTOR INTEGRATION AND INTEGRAL THEOREM 9 Hours

Line integrals over scalar and vector fields; work done and path independence; surface integrals and flux; Green's theorem in the plane and its applications; Stokes' theorem; Divergence theorem (Gauss's theorem); applications to area and volume computation; engineering applications in electromagnetism, fluid mechanics, and mass conservation.

AI Integration: Python SciPy integrate for numerical computation of line and surface integrals; 3D flux visualization with colour-coded normal flux using Matplotlib; numerical verification of Green’s and Stokes’ theorems; AI-assisted physical interpretation of the Divergence theorem in terms of source density; prompting AI to explain connections to Gauss’s law and student evaluation of AI explanations.

TEXT BOOKS:

1. E. Kreyszig, *Advanced Engineering Mathematics*, 11th Edition, International Adaptation. Hoboken, NJ, USA: Wiley, 2025.
2. B. S. Grewal, *Higher Engineering Mathematics*, 45th Edition, New Delhi, India: Khanna Publishers, 2024.
3. J. R. Hass, C. E. Heil, M. D. Weir, and P. Bogacki, *Thomas' Calculus*, 15th Edition, Pearson, 2022.

REFERENCE BOOKS:

1. R. Haberman and P. Choboter, *Applied Partial Differential Equations*, 6th Edition, Boston, MA, USA: Pearson, 2025.
2. D. G. Zill and W. S. Wright, *Advanced Engineering Mathematics*, 6th Edition, Burlington, MA, USA: Jones & Bartlett Learning, 2018.
3. I. N. Sneddon, *Elements of Partial Differential Equations*. Mineola, NY, USA: Dover Publications, 2006.
4. G. B. Arfken, H. J. Weber, and F. E. Harris, *Mathematical Methods for Physicists: A Comprehensive Guide*, 7th Edition, Academic Press (Elsevier), 2013, ISBN:978-0-12-384654-9.

Free/Open-Source AI Tools /Software:

1. Google Colab
2. Python (NumPy, SymPy, SciPy, Matplotlib)
3. Scilab
4. Sage Math
5. GeoGebra

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	2	–	–	–	–	–	–
CO2	3	3	3	3	2	–	–	–	–	–	–
CO3	3	3	3	3	2	–	–	–	–	–	–
CO4	3	3	3	3	2	–	–	–	–	–	–
CO5	3	3	3	3	2	–	–	–	–	–	–

Correlation Levels: 3- High; 2-Moderate; and 1- Low

I B.Tech. II Sem- Common to CSE(AI), AI, CSE (AI&ML), CSE (AI&R) and CSE (AI& DE)

L	T	P	C
3	0	2	4

26ACA01: ARTIFICIAL INTELLIGENCE

Course Outcomes (COs)

After successful completion of the course, students will be able to:

CO1 : Analyze AI environments using PEAS specifications and environment properties, design appropriate intelligent-agent architectures for given problem settings.

CO2 : Formulate real-world problems as state-space search problems and apply uninformed, informed, local search algorithms to obtain effective solutions.

CO3 : Represent knowledge using constraint-based, propositional and first-order logic formalisms and apply appropriate inference mechanisms to derive and validate conclusions.

CO4 : Formulate classical planning problems using PDDL and apply planning and probabilistic reasoning techniques, including Bayesian networks, to make decisions under uncertainty.

CO5 : Apply supervised learning techniques, including decision trees and linear models and evaluate predictive models using appropriate model-selection and validation techniques

UNIT– I: INTRODUCTION TO AI AND INTELLIGENT AGENTS 9 Hours

AI definitions, foundations (philosophy, mathematics, neuroscience, linguistics, computer science); history of AI-symbolic AI, expert systems, Intelligent agents: agents and environments; rationality; PEAS framework; environment properties-observable, deterministic, episodic, static, discrete, single/multi-agent. Agent types: simple reflex, model-based reflex, goal-based, utility-based, learning agents.

AI Integration: Use ChatGPT to apply the PEAS framework to one real-world AI systems; use Claude to compare capabilities and limitations of GPT-4, Claude, and Gemini; use Notebook LM to generate study notes.

UNIT– II: SOLVING PROBLEMS BY SEARCHING 9 Hours

Problem-solving agents; formulating problems: states, actions, transition model, goal test, path cost Example problems (toy and real-world) Uninformed search strategies: BFS, Uniform-Cost Search, DFS, Depth-Limited & Iterative Deepening Search, Bidirectional Search, Informed (Heuristic) search: Greedy Best-First Search, A* Search, heuristic functions, admissibility & consistency, Local search algorithms & optimization: Hill-

Climbing, Simulated Annealing, Local Beam Search.
AI Integration: Use ChatGPT to trace BFS, DFS, and A* on a state space graph; use Claude to explain Greedy Best-First Search, use Visu Algo or an AI assistant to visualize search algorithm execution.

UNIT– III: KNOWLEDGE REPRESENTATION 9 Hours

Constraint Satisfaction Problems: formulation, constraint propagation, backtracking search Logical Agents: knowledge-based agents, the Wumpus World, logic fundamentals Propositional Logic: syntax, semantics, inference, entailment Propositional theorem proving: resolution, Horn clauses, forward/backward chaining, First-Order Logic: syntax, semantics, quantifiers, knowledge engineering Inference in First-Order Logic: unification, forward chaining, backward chaining, resolution.
AI Integration: Use ChatGPT to trace forward and backward chaining on a knowledge base; use Claude to explain Propositional theorem proving and First-Order Logic.

UNIT– IV: PLANNING AND UNCERTAIN KNOWLEDGE 9 Hours

Automated Planning: PDDL (planning domain definition language), classical planning problem representation Planning algorithms: forward/backward state-space search, planning graphs, hierarchical planning (overview) Quantifying **Uncertainty**: acting under uncertainty, basic probability notation, axioms of probability, Bayes' Rule and its use; probabilistic reasoning Probabilistic Reasoning: Bayesian networks representation, semantics, construction.
AI Integration: Use ChatGPT to develop and validate simple PDDL planning problems, generate action sequences, and trace forward and backward state-space search for classical planning problems, Use Claude to explain and construct Bayesian networks.

UNIT – V: LEARNING FROM EXAMPLES 9 Hours

Forms of Learning: supervised, unsupervised, reinforcement learning - an overview Supervised Learning: learning from examples, task definition, hypothesis space Decision Tree learning: algorithm, choosing attribute tests, generalization and overfitting Model selection and evaluation: cross-validation, bias-variance tradeoff Linear Regression and Classification: linear models, gradient descent.
AI Integration: Use ChatGPT to explain and demonstrate the differences between supervised, unsupervised, and reinforcement learning, Use Claude to analyze model selection and evaluation concepts such as cross-validation, bias-variance trade-off, and overfitting.

LIST OF EXPERIMENTS:

Experiment-1: Write a python program to implement Pandas Series with labels.
Experiment-2: a Create a Pandas Series from a dictionary.
b Creating a Pandas Data Frame.
Experiment-3: Write a program which makes use of the following Pandas methods
i) describe () ii) head () iii) tail () iv) info ()
Experiment-4: Implementation of Intelligent Agents using PEAS Framework

Experiment-5: Write a Python program to implement a Simple Reflex Agent that selects actions based only on the current percept

Experiment-6: Write a Python program to Implement a Goal-Based Agent

Experiment-7: Write a Python program to implement Best First Searching Algorithm

Experiment-8: Write a Python program to implement Depth First Searching Algorithm

Experiment-9: Write a Python program to implement Heuristic Search

Experiment-10: Write a python program to implement A* algorithm.

Experiment-11: Write a Python program to implement Knowledge Representation using Semantic Networks and Frames

Experiment-12: Implementation of Decision Tree Learning with Model Evaluation and Cross-Validation

Experiment-13: Implementation of Linear Regression and Classification using Gradient Descent

TEXT BOOKS:

1. S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th edition. Pearson, 2022.

REFERENCE BOOKS:

1. G. F. Luger, *Artificial Intelligence: Structures and Strategies for Complex Problem Solving*, 6th Edition, Pearson, 2009
2. Ethem Alpaydin, *Introduction to Machine Learning*, 4th Edition, MIT Press, 2020.
3. I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

Software:

- Python 3.x; Jupyter Notebook / Google Colab; NumPy; Pandas; Matplotlib; scikit-learn

ONLINE RESOURCES:

NPTEL – Introduction to AI, IIT Madras – nptel.ac.in/courses/106106139

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	–	2	–	–	–	2	–	-	3	3
CO2	3	3	3	2	3	–	–	–	2	–	-	3	3
CO3	3	3	2	2	2	–	–	2	2	–	-	3	2
CO4	3	3	3	3	3	-	–	2	2	-	-	3	3
CO5	3	3	3	3	3	–	–	2	3	-	-	3	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B. Tech II Semester – Common to CSE, CSC, CSM, CAI, AI, IT, CSO, CSD, CSBS, AI&DS, CSE (QC), CSE (AI&R), CSE (CC&A), CSE (AI&DE)

L	T	P	C
3	0	0	3

26ACS04: COMPUTER ORGANIZATION AND ARCHITECTURE

Course Outcomes (COs)

After successful completion of the course, the student will be able to:

CO1 : Explain the functional units, register organization and instruction cycle of a computer, analyze instruction formats, addressing modes and RISC/CISC architectures.

CO2 : Analyze CPU organization, register-transfer operations and micro-operations, compare hardwired and microprogrammed control-unit designs.

CO3 : Apply computer arithmetic algorithms and floating-point representations to perform and analyze integer and floating-point arithmetic operations and ALU functions.

CO4 : Analyze memory hierarchy, cache organization and mapping/replacement techniques, virtual memory mechanisms to evaluate memory system performance.

CO5 : Evaluate I/O organization techniques, bus architectures, DMA and interrupt mechanisms, advanced processor organizations including pipelining, superscalar, multicore and parallel processing architecture.

UNIT-I: BASIC COMPUTER ORGANIZATION AND INSTRUCTION SET ARCHITECTURE 9 Hours

Functional units of a computer, Computer system organization and architecture, Registers and register organization, CPU organization, Instruction cycle, Fetch, decode and execute operations, Instruction formats, Instruction types, Addressing modes, Instruction Set Architecture (ISA), RISC and CISC architectures, Stack organization, Subroutine and procedure handling

AI Integration: Use ChatGPT or Gemini to compare and contrast code density and execution efficiency between RISC and CISC architectures for a specific algorithm.

UNIT-II: CPU ORGANIZATION AND CONTROL UNIT 9 Hours

Central Processing Unit organization, Register Transfer Language (RTL), Register transfer and micro-operations, Arithmetic, logical and shift micro-operations, Instruction execution, Hardwired control unit, Microprogrammed control unit, Control memory, Microinstructions, Horizontal and vertical microprogramming, Control unit design.

AI Integration: Use GitHub Copilot to generate Register Transfer Language (RTL)

descriptions for basic arithmetic and logic micro-operations.

UNIT-III: COMPUTER ARITHMETIC 9 Hours

Integer representation, Signed and unsigned numbers, Fixed-point arithmetic, Floating-point representation, Floating-point arithmetic, Addition and subtraction algorithms, Multiplication algorithms- Booth's algorithm, Array multiplication; Division algorithms- Restoring division, Non-restoring division; Arithmetic Logic Unit (ALU) organization.

AI Integration: Use an AI/LLM tool to explain 2's complement addition/subtraction with overflow detection and walk through Carry- Lookahead Adder design, comparing delay versus ripple carry adders.

UNIT- IV: MEMORY ORGANIZATION 9 Hours

Memory hierarchy, Semiconductor memory, RAM and ROM, Cache memory, Cache organization and mapping techniques - Direct mapping, Associative mapping, Set-associative mapping; Cache replacement policies, Cache write policies, Virtual memory, Paging, Page replacement, Translation Lookaside Buffer (TLB), Memory management, Secondary storage, RAID.

AI Integration: Use an AI/LLM tool to explain memory hierarchy and cache operation (hit, miss, LRU/FIFO) with a direct-mapped cache example, calculating hit rate manually.

UNIT-V: INPUT/OUTPUT ORGANIZATION AND ADVANCED PROCESSOR ORGANIZATION 9 Hours

I/O interface, I/O ports, Programmed I/O, Interrupt-driven I/O, Direct Memory Access (DMA), I/O processors, Interrupt priority, Bus organization, System bus, Synchronous and asynchronous data transfer, Parallel and serial communication, Pipelining, Instruction pipeline, Pipeline hazards , Data, structural and control hazards, Superscalar processors, Introduction to multicore processors , Introduction to parallel processing.

AI Integration: Use an AI/LLM tool to compare programmed I/O, interrupt-driven I/O, and DMA in terms of CPU involvement and throughput, and trace the complete interrupt handling sequence.

TEXT BOOKS

1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky and Naraig Manjikian, *Computer Organization and Embedded Systems*, 6th Edition, McGraw-Hill Education, 2023.

2. David A. Patterson and John L. Hennessy, *Computer Organization and Design: The Hardware/Software Interface, MIPS Edition*, 6th Edition, Morgan Kaufmann/Elsevier, 2020.

REFERENCE BOOKS

1. W. Stallings, *Computer Organization and Architecture:* Designing for Performance, 11th Edition., Pearson, 2019.

2. S. R. Sarangi, *Advanced Computer Organization & Architecture*, 1st Edition, McGraw-Hill Education, 2021.

3. A. S. Tanenbaum and T. Austin, *Structured Computer Organization*, 6th Edition, Pearson, 2016.

ONLINE RESOURCES:

1. NPTEL – Computer Architecture
2. MIT OpenCourseWare – Computation Structures / Computer Architecture
3. GeeksforGeeks – Computer Organization and Architecture
4. Wikibooks – Computer Systems
5. Coursera – Computer Architecture Courses
6. YouTube – NPTEL Computer Architecture Lectures
7. Patterson & Hennessy resources

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	2	2	-	-	-	-	-	-	3	2
CO2	3	3	3	2	2	-	-	-	-	-	-	3	2
CO3	3	3	2	3	3	-	-	-	-	-	-	3	2
CO4	3	3	3	3	3	-	-	-	-	-	-	3	3
CO5	3	3	3	3	3	-	-	-	-	-	-	3	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B.Tech. I Sem- Common to CSC, AI, CSE(AI), CSE(QC), AI&DS, IT, CSO and CSE (CCA)

I B.Tech. II Sem- Common to CSE, CSE(AI&ML), CSE(AI&DE), CSE(DS), CSBS and CSE(AI&R)

L	T	P	C
3	0	0	3

26AEC03: DIGITAL LOGIC DESIGN

Course Outcomes (COs)

After successful completion of the course, students will be able to:

CO1 : Apply number systems, binary arithmetic and Boolean algebra to represent, analyze and simplify digital logic expressions and functions.

CO2 : Minimize and implement optimized Boolean functions using Karnaugh maps, SOP/POS forms, don't-care conditions and universal gates.

CO3 : Analyze and design combinational circuits including adders, subtractors, multipliers, comparators, multiplexers, decoders, encoders and code converters for specified applications.

CO4 : Analyze and design sequential circuits using flip-flops, registers and counters, including state reduction, state assignment and timing behaviour.

CO5 : Analyze memory devices and programmable logic devices, and implement switching functions using ROM, PROM, PLA and PAL with appropriate error detection and correction techniques.

UNIT – I: NUMBER SYSTEM & BOOLEAN ALGEBRA 9 Hours

Digital systems, Binary Numbers, Octal Numbers, Hexadecimal Numbers, Number base conversions, complements of numbers, Signed Binary numbers, Binary Arithmetic: addition, subtraction, multiplication, division Binary codes. Boolean algebra – Basic definition, Basic theorems and properties, Boolean Functions, Canonical & Standard forms, other logic operations & Digital logic gates.

AI Integration: Students use AI/LLM tools to solve and verify number conversions and signed binary arithmetic problems. A Python (Google Colab) template is used to convert decimal to binary/octal/hex and compare canonical SOP, POS, and simplified Boolean expressions.

Free/Open-Source AI Tools: Python (NumPy, SymPy) on Google Colab; Logisim; online Boolean calculators; ChatGPT / Gemini

UNIT – II: GATE LEVEL MINIMIZATION 9 Hours

The map method, four variable K-map, five variable K-map, POS & SOP Simplification, Don’t care conditions, NAND & NOR Implementation, other two-level Implementations, Exclusive-OR Function.

AI Integration: Students use AI/LLM tools to NAND/NOR-only implementations through worked examples. A Python K-map template auto-groups minterms and verifies minimized SOP expressions against manual solutions.

Free/Open-Source AI Tools: Python (NumPy) on Google Colab; Logisim; K-map solver tools online; ChatGPT / Gemini.

UNIT – III: COMBINATIONAL CIRCUITS 9 Hours

Combinational circuits, Analysis & Design procedure, Binary Adder and Subtractor, Decimal Adder, Binary Multiplier, Magnitude comparator, Decoder, Encoders, Multiplexers, De-multiplexers, Code Converters, priority encoders.

AI Integration: Students use AI/LLM tools to derive adder and BCD adder logic and to write and trace a Verilog program for a BCD-to-Excess-3 converter. A 4-to-1 MUX and 2-to-4 decoder are simulated in Python (Google Colab) to verify Boolean function implementation.

Free/Open-Source AI Tools: Python (NumPy) on Google Colab; Logisim; EDA Playground (free Verilog simulation); ChatGPT / Gemini.

UNIT – IV: SEQUENTIAL CIRCUITS 9 Hours

Sequential Circuits, Latches, Flips-Flops, Conversion of Flip Flops, Analysis of Clocked sequential circuits, State Reduction & Assignment, Design procedure, Registers & Counters –Registers, Shift Registers, Ripple Counters, Synchronous counters, Design of modulo-N Counters, Ring and Johnson Counters.

AI Integration: Students use AI/LLM tools to work through flip-flop excitation tables and JK-to-D conversion, then write a Verilog behavioural model with test bench for a JK flip-flop and ring counter. A 4-bit synchronous up-counter and SIPO shift register are simulated in Python to verify state sequences.

Free/Open-Source AI Tools: Python (NumPy) on Google Colab; Logisim; EDA Playground (free Verilog simulation); ChatGPT / Gemini.

UNIT – V: MEMORIES 9 Hours

Random - Access Memory, Memory Decoding, Error Detection and correction, Read - Only Memory, Programmable Logic Array (PLA), Programmable Array Logic (PAL), Sequential Programmable Devices, Realization of Switching Functions Using PROM, PAL and PLA.

AI Integration: Students use AI/LLM tools to design and compare PLA, PAL, PROM, and FPGA architectures.

Free/Open-Source AI Tools: Python on Google Colab; EDA Playground; Intel Quartus Prime Lite / Xilinx Vivado (free);

TEXT BOOKS:

1. Mano M.M. and Ciletti M.D., *Digital Design*, 6thEdition, Pearson Education, 2019.
2. Roth C.H, Kinney L.L, Eugene B.John., *Fundamentals of Logic Design*, Enhanced 7thEdition, Cengage Learning, 2024.

REFERENCE BOOKS

1. C.V.S Rao., "*Switching Theory and Logic Design"*, 3rd Edition, Pearson Education, 2009
2. R.P Jain and Kishor Sarawadekar., "*Modern Digital Electronics"*, 5th Edition, Tata McGraw Hill, 2022
3. Donald.P. Leach, Albert Paul Malvino, Goutam Saha., "*Digital Principles and Applications*", 8th Edition, TMH, 2018

Course Articulation Matrix:

Course outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	2	1	2	–	–	–	–	–	–	3	3	2
CO2	3	3	3	2	3	–	–	–	–	–	–	3	3	2
CO3	3	3	3	2	3	–	–	–	–	–	–	3	3	2
CO4	3	3	3	3	3	–	–	–	–	–	–	3	3	2
CO5	3	3	3	3	3	–	–	–	–	–	–	3	3	2

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B.Tech. II Sem- Common to CSE(AI), AI, CSE (AI&ML), CSE (AI&R) and CSE (AI& DE)

L	T	P	C
0	0	3	1.5

26AAI02: PROMPT ENGINEERING

COURSE OUTCOMES (COs)

After successful completion of the course, students will be able to:

CO1 : Explain the fundamental concepts, terminology and evolution of prompt engineering and distinguish prompting approaches from feature engineering and model fine-tuning.

CO2 : Design and refine closed-ended and open-ended prompts for diverse AI application tasks by balancing output control, structure, creativity and flexibility.

CO3 : Develop systematic probing and prompt-optimization workflows to identify model limitations, compare prompt variants, improve task-specific prompt performance using evaluation harnesses.

CO4 : Evaluate prompt performance using quantitative metrics and human-in-the-loop rubrics, conduct bias audits and mitigation procedures to assess the reliability, fairness and suitability of prompt-based systems for deployment.

CO5 : Apply and critically evaluate prompt-engineering strategies to solve authentic problems across mathematics, science, programming and engineering domains, analyze their effectiveness, limitations and deployment considerations.

LIST OF EXPERIMENTS:

Experiment- 1: Foundations of Prompt Engineering

a) Overview of prompt engineering and its significance in AI, including how it differs from traditional feature engineering and model fine-tuning.

b) Historical context and evolution of prompt-based approaches, from early template-based AI systems through GPT-style few-shot learning to modern instruction-tuned and agentic models.

c) Core terminology and mental models - tokens, context windows, zero-shot vs. few-shot vs. chain-of-thought prompting - establishing a shared vocabulary for the rest of the course.

d) Hands-on warm-up exercise: writing and comparing three prompt variations for the same task to observe how small wording changes affect output quality.

AI Integration: Use ChatGPT to demonstrate zero-shot, few-shot, and structured prompting for the same task, Use Claude to explore the evolution of prompt-based AI, from template-based systems to few-shot learning, instruction-tuned models, and modern agentic AI Prompt Comparison Activity**:** Students provide the same task to ChatGPT and Claude using

three prompt variations-basic, few-shot, and carefully structured-and compare the resulting outputs.

Experiment- 2: Closed-Ended Prompts

a) Understanding and creating prompts for specific answers, including binary, multiple-choice, and constrained-output formats.
b) Applications in question-answering systems, structured data extraction, and classification tasks.
c) Techniques for enforcing output structure to make closed-ended prompts production-ready.
d) Common failure modes - ambiguous phrasing, leading questions, and answer-choice bias - with strategies to avoid them.

AI Integration: Use ChatGPT to design and test binary, multiple-choice, and constrained-output prompts for question-answering, classification, and structured information extraction tasks. Use Claude to generate structured outputs such as JSON, key-value pairs, tables, and predefined labels**,** Use AI tools to identify and demonstrate common prompting failures such as ambiguous instructions, leading questions, answer-choice bias, unwanted explanations, and format violations.

Experiment- 3: Open-Ended Prompts

a) Crafting prompts for creative responses, brainstorming, and exploratory tasks.
b) Applications in language generation models, including storytelling, summarization, and ideation tools.
c) Balancing creativity and coherence-using temperature, constraints, and guiding examples to shape open-ended output without over-restricting it.
d) Case comparison of open- vs. closed-ended prompting on the same task to illustrate trade-offs in control vs. flexibility.

AI Integration: Use ChatGPT to create and experiment with open-ended prompts for brainstorming, storytelling, summarization, creative writing, Use Claude to explore creative ideation and long-form generation, testing different prompt structures and guiding examples Use ChatGPT and Claude to solve the same task using both open-ended and closed-ended prompts.

Experiment- 4: Probing Prompts

a) Designing prompts to reveal model biases, blind spots, and reasoning gaps.
b) Ethical considerations in using probing prompts, including consent, transparency, and responsible disclosure of findings.

AI Integration: Use ChatGPT to design and test probing prompts that examine model biases, blind spots, inconsistencies, hallucinations, and reasoning gaps. Use Claude to conduct structured model evaluation, identifying potential biases, inconsistencies, and limitations in AI-generated responses. Use AI tools to help students develop a structured probing process.

Experiment-5: Systematic Probing Framework

a) Frameworks for systematic probing
b) Documenting and reporting probing results in a way that supports model

AI Integration: Use ChatGPT to design a systematic probing framework by defining evaluation objectives, generating controlled prompt variations, categorizing model responses, Use Claude to assist in analyzing and interpreting probing results, including identifying recurring patterns

Experiment- 6: Optimizing Prompt Selection

a) Methods for selecting optimal prompts for specific tasks along with automated prompt search.
b) Customizing prompts based on model behavior and observed output patterns.
c) Building a lightweight evaluation harness to compare prompt variants at scale.

AI Integration: Use ChatGPT to generate multiple prompt variants for a given task, refine prompts based on observed outputs, Use Claude to analyze model responses, identify recurring errors and output patterns, and recommend prompt modifications to improve accuracy

Experiment-7: Evaluating Prompt Performance

a) Metrics and methodologies for assessing model performance with prompts, including accuracy, relevance, coherence, and task-specific scoring.
b) Interpreting and analyzing results to identify systematic weaknesses.

AI Integration: Use ChatGPT to evaluate prompt outputs against predefined criteria such as accuracy, relevance, coherence, completeness, Use Claude as a secondary evaluator to independently assess the same outputs. Use AI tools to analyze evaluation results and identify recurring weaknesses such as hallucinations, irrelevant responses, incomplete answers

Experiment- 8: Evaluating Prompt Performance

a) Introduction to human-in-the-loop evaluation and inter-rater reliability for subjective tasks.
b) Building a repeatable evaluation rubric that can be reused across projects and teams.

AI Integration: Use ChatGPT to assist in designing human-in-the-loop evaluation criteria and generate sample responses for students to evaluate for accuracy, relevance, Use Claude as an AI-assisted evaluator to independently score the same responses using a predefined rubric.

Experiment- 9: Bias Mitigation in Prompt Engineering

a) Strategies to identify and address biases introduced by prompts, including wording, framing, and example-selection bias.
b) Ensuring fairness and inclusivity in prompt-based models through diverse test sets and representative examples.

AI Integration: Use ChatGPT to design and test prompt variations that help identify wording bias, framing bias, and example-selection bias, Use Claude to review prompts and test datasets for potential fairness, inclusivity, stereotyping

Experiment- 10: Bias Mitigation in Prompt Engineering

a) Practical bias-auditing checklist applicable before deploying any prompt-based system.
b) Case discussion of a real-world bias incident and the prompting changes used to mitigate it.

AI Integration: Use ChatGPT to create and apply a pre-deployment bias-auditing checklist for prompt-based systems, Use Claude to independently review the same prompt and test cases for potential bias, stereotyping, exclusion, or inconsistent treatment. Use AI tools to generate a structured report containing audit criteria, test cases, observed outputs, identified risks

Experiment- 11: Case Studies in Prompt Engineering

a) Algorithm Design - Breaking Down Problem-Solving Steps: A student is asked to write a function that finds the second-largest number in a list, without using built-in sort.
b) A tutoring platform uses an AI to help students solve algebra word problems.

AI Integration: Use ChatGPT to help students develop prompts that guide an AI through the steps for finding the second-largest number in a list without using built-in sorting functions. Use Claude to review the generated Python solutions, identify logical errors, edge cases, and unnecessary operations**,**

Experiment- 12: Case Studies in Prompt Engineering

a) Conceptual Explanation vs. Formula Dumping: A high-school physics app wants to explain why objects in free fall accelerate at the same rate regardless of mass.
b) Balancing Equations and Safety Context: A chemistry homework helper needs to balance a combustion reaction and explain the process.

AI Integration: Use ChatGPT to generate explanations of free-fall acceleration at different levels, such as school-level, intuitive, and technical explanations. Use Claude to rewrite the same physics explanation for different audiences and identify whether the response explains

TEXT BOOKS:

1. Ian Khan, *The Quick Guide to Prompt Engineering: Generative AI Tips and Tricks for ChatGPT, Bard, DALL-E, and Midjourney: An Indian Adaptation*, Wiley India, 1st Edition, 2024.

REFERENCE BOOKS:

1. Berryman and A. Ziegler, *Prompt Engineering for LLMs: The Art and Science of Building Large Language Model-Based Applications*, O'Reilly Media, 1st Edition, 2024.

2. James Phoenix and Mike Taylor, *Prompt Engineering for Generative AI: Future-Proof Inputs for Reliable AI Outputs at Scale*, O'Reilly Media, 1st Edition, 2024.

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	1	–	–	3	–	–	–	1	–	–	3	2
CO2	1	2	2	1	3	–	–	–	3	–	–	3	3
CO3	2	3	1	3	3	–	–	–	2	–	–	3	3
CO4	2	2	3	2	3	–	–	1	2	–	–	3	3
CO5	2	2	3	2	3	–	–	2	3	–	–	3	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B.Tech. II Sem- Common to All Branches

L	T	P	C
0	0	1	0.5

26AHS09 - NSS/NCC /SCOUTS AND GUIDES

Course Outcomes (COs)

After the completion of the course the student will be able to

CO1 : Demonstrate discipline, self-confidence, responsibility and commitment to the values of service and civic responsibility.

CO2 : Identify and analyse basic needs and problems of local communities through observation, surveys and interaction with community stakeholders.

CO3 : Participate in planning and implementing community-service activities addressing identified social, health and environmental concerns.

CO4 : Demonstrate teamwork, leadership, communication and interpersonal skills while organizing and executing community-oriented activities.

CO5 : Promote environmental awareness and sustainable practices through activities related to waste management, recycling, pollution prevention and sustainable living.

CO6 : Demonstrate ethical conduct, empathy and social responsibility towards fellow citizens and disadvantaged sections of society.

UNIT – I: ORIENTATION

General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, career guidance.

ACTIVITIES:

i) Conducting –ice breaking sessions-expectations from the course-knowing personal talents and skills
ii) Conducting orientations programs for the students –future plans-activities-releasing road map etc.
iii) Displaying success stories-motivational biopics- award winning movies on societal issues etc.
iv) Conducting talent show in singing patriotic songs-paintings- any other contribution.

UNIT – II: NATURE & CARE ACTIVITIES

i) Best out of waste competition.
ii) Poster and signs making competition to spread environmental awareness.
iii) Recycling and environmental pollution article writing competition.
iv) Organising Zero-waste day.
v) Digital Environmental awareness activity via various social media platforms.

vi) Virtual demonstration of different eco-friendly approaches for sustainable living.
vii) Write a summary on any book related to environmental issues.

UNIT – III: COMMUNITY SERVICE ACTIVITIES

i) Conducting One Day Special Camp in a village contacting village-area leaders-Survey in the village, identification of problems- helping them to solve via media-authorities-experts-etc.
ii) Conducting awareness programs on Health-related issues such as General Health, Mental health, Spiritual Health, HIV/AIDS,
iii) Conducting consumer Awareness. Explaining various legal provisions etc.
iv) Women Empowerment Programmes- Sexual Abuse, Adolescent Health and Population Education.
v) Any other programmes in collaboration with local charities, NGOs etc.

TEXT BOOKS:

1. S. K. Mangal and S. Mangal, *Health and Physical Education*. New Delhi, India: PHI Learning Pvt. Ltd., 2021.
2. W. J. Broad, *The Science of Yoga*. New York, NY, USA: Simon & Schuster, 2012.

REFERENCE BOOKS:

1. Nirmalya Kumar Sinha & Surajit Majumder, *A Text Book of National Service Scheme* Vol; I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6)
2. *Red Book - National Cadet Corps* – Standing Instructions Vol I & II, Directorate General of NCC, Ministry of Defence, New Delhi
3. Davis M. L. and Cornwell D. A., "Introduction to Environmental Engineering", McGraw Hill, New York 4/e 2008
4. Masters G. M., Joseph K. and Nagendran R. "Introduction to Environmental Engineering and Science", Pearson Education, New Delhi. 2/e 2007
5. Ram Ahuja. *Social Problems in India*, Rawat Publications, New Delhi.

GENERAL GUIDELINES:

1. Institutes must assign slots in the Timetable for the activities.
2. Institutes are required to provide instructor to mentor the students.

EVALUATION GUIDELINES:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit. Each activity shall be evaluated by the concerned teacher for 15 marks, totalling to 90 marks.
- A student shall be evaluated by the concerned teacher for 10 marks by conducting viva voce on the subject.

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	–	–	–	–	2	3	3	2	1	2
CO2	1	3	1	1	–	3	2	2	3	2	2
CO3	1	2	2	1	1	3	3	3	3	2	2
CO4	1	–	1	–	–	2	–	3	3	3	2
CO5	1	2	2	1	1	3	3	2	2	2	3
CO6	1	–	–	–	–	3	3	3	3	2	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low